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Correlation between the PFAS levels in bovine blood and the levels in the meat and offal of these animals

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Abstract:
In 2023, the slaughterhouse-sampled meat and liver of some cows from Belgian businesses were found not to comply with the maximum levels for PFOS described in European legislation. These post-mortem results led to a ban on the entry into the food chain of food obtained from animals coming from the affected businesses.

The Federal Agency for the Safety of the Food Chain (FASFC) wishes to propose tools to livestock farmers to enable them to manage PFAS contamination and take the best possible decisions and this within the framework of their self-checking system. Therefore, the Scientific Committee (SciCom) established at the FASFC assessed the possibility of establishing a guide value for PFAS compounds in bovine blood to estimate whether PFAS levels in meat, liver and kidney comply with the European maximum levels (Regulation (EU) 2023/915). To this end, the SciCom relied on (i) Danish Decree No 1386 of 29/11/2023, (ii) literature data and (iii) monitoring data from the FASFC. In addition, a linear regression analysis was applied to data of PFOS concentrations in blood plasma and muscle tissue from the same animals (paired results for 28 cows) (Johnston *et al.*, 2023). This analysis performed on a logarithmic transformation of the raw data allowed to determine that a PFOS content of 0.3 µg/kg in meat (= legal European maximum content) corresponds to a most probable PFOS content of 6.2 µg/L in the blood plasma with a 95% confidence interval of 2.5 to 23.4 µg/L. For liver and kidney, it was not possible to apply this type of approach due to the lack of available paired data (PFOS content in blood and tissues from the same animal).

The obtained lower limit of 2.5 µg/L is close to, but more conservative, than the guide value of 3.3 µg PFOS/L blood plasma applied by the Danish authorities. The SciCom considers that a PFOS level of 2.5 µg/L blood plasma is an appropriate guide value to estimate a compliance of a PFOS level in the meat, liver and kidney. For PFOA, PFHxS and PFNA, there is less to no data to derive such a guide value in blood plasma. Based on the available data, the SciCom considers that the same concentration of 2.5 µg/L blood plasma can also be applied to estimate a compliance of levels of PFOA, PFHxS or PFNA in bovine meat, liver and kidney. It should be stressed that this guide value of 2.5 µg/L is subject to significant uncertainties.

Ratios between PFAS levels in bovine muscle tissue, liver and kidney were assessed using (i) FASFC control data and (ii) data from the scientific literature. The aim was to assess whether the liver or kidneys of a slaughtered bovine should be destroyed if the meat is non-compliant or vice versa. On the basis of available results for the liver (n=16), it could be determined that the PFOS concentration in the liver is between 6 and 34 times higher (on average 21 times) than that in muscle tissue, while the PFOS concentration in the kidneys (n=8) is between 3 and 12 times higher (on average 8 times) than that in muscle tissue. No such ratio could be derived for PFOA, PFNA and PFHxS due to insufficient data. It is noted that these ratios for PFOS were determined from a relatively limited number of data from different studies with differences in experimental design, and that no distinction was made between breed, sex and lactation status of the cattle when deriving the ratios.

Finally, the SciCom evaluated the possibility of determining, based on ante-mortem blood analysis, a withdrawal time after which bovine muscle tissue, liver and kidney will be compliant for PFAS after removal of the source of contamination. For this purpose, literature sources regarding the depletion kinetics of PFAS in bovine blood were reviewed. For PFOS, elimination half-lives in blood of 39 (lactating cow) to 120 days (steer) were distinguished.

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For PFOA, elimination half-lives in blood were 1.3 (lactating cow) and 19.2 days (steers). For PFNA the elimination half-lives in blood were 8.7 (lactating cow) and 12.3 days (beef cattle), and for PFHxS the half-life was 9.3 days for beef cattle.

Chou *et al.* (2023) implemented a physiological-based pharmacokinetic (PBPK) model implemented in an interactive generic platform publicly accessible via the internet where tissue withdrawal times for PFOA, PFOS and PFHxS can be calculated for beef cattle and dairy cows. This platform could be used to predict withdrawal times for these PFAS in beef cattle and dairy cows. An important limitation of the platform is that several parameter values need to be known to enter into the application, in particular the number of animals, PFAS concentration in soil or water, time interval of dosing of PFAS contamination, exposure frequency and exposure duration. It will often prove difficult if not impossible for a livestock farmer to know or determine these values. In addition, the source of contamination may be different, it could be, for example, contaminated feed, and not soil or water. In addition, scientific publications report other equations that would approximate depletion kinetics in blood plasma or blood serum (Johnston *et al.*, 2023; Mikkonen *et al.*, 2023), but also these turn out to be of little use in practice because of the required parameters that need to be known and the complexity. Therefore, the working group itself developed a pragmatic approach for calculating a withdrawal time after which a second blood sample should be collected and analysed, in case of an initial blood test indicating an excess PFAS level. The equation for calculating such a withdrawal time (see recommendations below) contains as constants the target value for PFAS in blood plasma of 2.5 µg/L and an elimination half-life for PFAS (for which a value from literature should be selected). Therefore, such a calculated withdrawal time is subject to significant uncertainties.

When a contamination by PFAS is detected on a farm, it is recommended to identify and eliminate the source of contamination (e.g. water, soil or feed) or to prevent animals from having access to it. The SciCom recommends the threshold value of 2.5 µg/L of bovine blood plasma as a suitable guide value to estimate ante-mortem a compliance against the legal European maximum levels of PFOS, PFOA, PFHxS or PFNA in the meat, liver and kidney. As discussed above, this guide value of 2.5 µg/L is subject to significant uncertainties. The SciCom emphasises that it cannot be excluded that possible non-compliant results of PFAS may be observed in the meat, liver or kidney of cattle following a blood test complying with the 2.5 µg/L guide value.

The SciCom therefore recommends assessing in the near future the number of cases in which a level of PFOS, PFOA, PFHxS or PFNA in blood plasma not exceeding the guide value of 2.5 µg/L is actually related to meat, kidney and liver conformity for these 4 PFAS. It is therefore recommended to determine and to collect paired data (from the same animal) concerning PFAS in the blood plasma, muscle tissue, liver and kidney of bovines originating from businesses affected by a PFAS contamination. These additional data can then serve as a basis for confirming or, if necessary, adjusting the guide value of 2.5 µg/L.

If an analysis of a bovine blood plasma would show that the concentration of PFOS, PFOA, PFHxS or PFNA is higher than the guide value of 2.5 µg/L, then it is recommended to perform another blood plasma analysis at a later date until a result lower than this value is obtained. This later data can be calculated using the following equation:

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$$withdrawal\ time\ t = \frac{(\log C_0 - \log 2{,}5) \times 2.303}{(0{,}693/DT_{50})}$$

Equation of the withdrawal time (t, in days) in blood plasma/serum to obtain the targeted guide value, where C_0 represents the starting concentration (µg/L) of PFOS, PFOA, PFHxS or PFNA in the blood plasma/serum, i.e. the concentration measured at initial sampling; 2,5 µg/L the suggested blood plasma/serum concentration guide value; DT_{50} the blood plasma/serum elimination half-life of PFOS, PFOA, PFHxS or PFNA (days).

Here, the SciCom also refers to the uncertainties discussed above. The withdrawal time should therefore be considered as an indication to determine a time when a second blood test may be useful. However, it cannot be excluded that after the estimated withdrawal time a bovine would still show a PFAS level in blood plasma higher than 2.5 µg/L. In addition, calculated withdrawal times are expected to be rather long based on the relatively long elimination half-lives of PFAS described in the literature. For these reasons, the practical usefulness of these withdrawal times is rather low.

In addition, for PFOS, it is recommended to use the following ratios to estimate its relative concentration in bovine muscle tissue, liver or kidney without having to carry out separate measurements in all these tissues: the PFOS concentration in liver is on average 21 times (between 6 and 34 times) higher than that in muscle tissue, while the PFOS concentration in kidney is on average 8 times (between 3 and 12 times) higher than that in muscle tissue. Significant uncertainties are applicable to these ratios. The SciCom recommends sampling paired samples (from the same animal) of muscle tissue, liver and kidney from Belgian bovines in the near future to verify these PFOS ratios between the different animal tissues.

Keywords: PFAS, cattle, bovines, blood, offal, meat, muscle, liver, kidney

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SciCom

SCIENTIFIC COMMITTEE
established at the Federal Agency for the Safety of the Food Chain

OPINION 01-2025

Regarding:

Correlation between the PFAS levels in bovine blood and the levels in the meat and offal of these animals

(SciCom 2024/10)

Scientific opinion adopted by the Scientific Committee on 16 January 2025

Keywords:
PFAS, cattle, bovines, blood, offal, meat, muscle, liver, kidney

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Summary

Correlation between the PFAS levels in bovine blood and the levels in the meat and offal of these animals

Background & Terms of reference

In 2023, the slaughterhouse-sampled meat and liver of some cows from Belgian businesses were found not to comply with the maximum levels for PFOS described in European legislation. These post-mortem results led to a ban on the entry into the food chain of food obtained from animals coming from the affected businesses.

The Federal Agency for the Safety of the Food Chain (FASFC) wishes to propose tools to livestock farmers to enable them to manage PFAS contamination and take the best possible decisions and this within the framework of their self-checking system.

Three questions are therefore addressed to the Scientific Committee:

(1) From what values of PFAS compounds (PFOS, PFOA, PFNA and PFHxS) in the blood of a bovine animal could one suspect a non-conformity (according to Regulation (EU) 2023/915) for the meat or offal (e.g. liver and kidney) originating from that animal?
(2) What are the correlations between the PFAS levels in bovine muscle tissue, liver and kidney?
(3) What are the depletion kinetics of PFAS (PFOS, PFOA, PFNA and PFHxS) in the blood of live cattle once the sources of PFAS contamination are removed?

Methodology

The opinion relies on literature review and FASFC control data, expert opinion and regression analysis of a selected dataset.

Conclusions

The Scientific Committee assessed the possibility of establishing a guide value for PFAS compounds in bovine blood to estimate whether PFAS levels in meat, liver and kidney comply with the European maximum levels (Regulation (EU) 2023/915). To this end, the SciCom relied on (i) Danish Decree No 1386 of 29/11/2023, (ii) literature data and (iii) monitoring data from the FASFC. In addition, a linear regression analysis was applied to data of PFOS concentrations in blood plasma and muscle tissue from the same animals (paired results for 28 cows) (Johnston *et al.*, 2023). This analysis performed on a logarithmic transformation of the raw data allowed to determine that a PFOS content of 0.3 µg/kg in meat (= legal European maximum content) corresponds to a most probable PFOS content of 6.2 µg/L in the blood plasma with a 95% confidence interval of 2.5 to 23.4 µg/L. For liver and kidney, it was not possible to apply this type of approach due to the lack of available paired data (PFOS content in blood and tissues from the same animal).

The obtained lower limit of 2.5 µg/L is close to, but more conservative, than the guide value of 3.3 µg PFOS/L blood plasma applied by the Danish authorities. The Scientific Committee considers that a PFOS level of 2.5 µg/L blood plasma is an appropriate guide value to estimate a compliance of a PFOS level in the meat, liver and kidney. For PFOA, PFHxS and PFNA, there is less to no data to derive such a guide value in blood plasma. Based on the available data, the Scientific Committee considers that the same concentration of 2.5 µg/L blood plasma can also be applied to estimate a compliance of levels of PFOA, PFHxS or PFNA in bovine meat, liver and kidney. It should be stressed that this guide value of 2.5 µg/L is subject to significant uncertainties.

Ratios between PFAS levels in bovine muscle tissue, liver and kidney were assessed using (i) FASFC control data and (ii) data from the scientific literature. The aim was to assess whether the liver or

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kidneys of a slaughtered bovine should be destroyed if the meat is non-compliant or vice versa. On the basis of available results for the liver (n=16), it could be determined that the PFOS concentration in the liver is between 6 and 34 times higher (on average 21 times) than that in muscle tissue, while the PFOS concentration in the kidneys (n=8) is between 3 and 12 times higher (on average 8 times) than that in muscle tissue. No such ratio could be derived for PFOA, PFNA and PFHxS due to insufficient data. It is noted that these ratios for PFOS were determined from a relatively limited number of data from different studies with differences in experimental design, and that no distinction was made between breed, sex and lactation status of the cattle when deriving the ratios.

Finally, the SciCom evaluated the possibility of determining, based on ante-mortem blood analysis, a withdrawal time after which bovine muscle tissue, liver and kidney will be compliant for PFAS after removal of the source of contamination. For this purpose, literature sources regarding the depletion kinetics of PFAS in bovine blood were reviewed. For PFOS, elimination half-lives in blood of 39 (lactating cow) to 120 days (steer) were distinguished. For PFOA, elimination half-lives in blood were 1.3 (lactating cow) and 19.2 days (steers). For PFNA the elimination half-lives in blood were 8.7 (lactating cow) and 12.3 days (beef cattle), and for PFHxS the half-life was 9.3 days for beef cattle.

Chou *et al.* (2023) implemented a physiological-based pharmacokinetic (PBPK) model implemented in an interactive generic platform publicly accessible via the internet where tissue withdrawal times for PFOA, PFOS and PFHxS can be calculated for beef cattle and dairy cows. This platform could be used to predict withdrawal times for these PFAS in beef cattle and dairy cows. An important limitation of the platform is that several parameter values need to be known to enter into the application, in particular the number of animals, PFAS concentration in soil or water, time interval of dosing of PFAS contamination, exposure frequency and exposure duration. It will often prove difficult if not impossible for a livestock farmer to know or determine these values. In addition, the source of contamination may be different, it could be, for example, contaminated feed, and not soil or water. In addition, scientific publications report other equations that would approximate depletion kinetics in blood plasma or blood serum (Johnston *et al.*, 2023; Mikkonen *et al.*, 2023), but also these turn out to be of little use in practice because of the required parameters that need to be known and the complexity. Therefore, the working group itself developed a pragmatic approach for calculating a withdrawal time after which a second blood sample should be collected and analysed, in case of an initial blood test indicating an excess PFAS level. The equation for calculating such a withdrawal time (see recommendations below) contains as constants the target value for PFAS in blood plasma of 2.5 µg/L and an elimination half-life for PFAS (for which a value from literature should be selected). Therefore, such a calculated withdrawal time is subject to significant uncertainties.

Recommendations

When a contamination by PFAS is detected on a farm, it is recommended to identify and eliminate the source of contamination (e.g. water, soil or feed) or to prevent animals from having access to it. The Scientific Committee recommends the threshold value of 2.5 µg/L of bovine blood plasma as a suitable guide value to estimate ante-mortem a compliance against the legal European maximum levels of PFOS, PFOA, PFHxS or PFNA in the meat, liver and kidney. As discussed above, this guide value of 2.5 µg/L is subject to significant uncertainties. The Scientific Committee emphasises that it cannot be excluded that possible non-compliant results of PFAS may be observed in the meat, liver or kidney of cattle following a blood test complying with the 2.5 µg/L guide value.

The Scientific Committee therefore recommends assessing in the near future the number of cases in which a level of PFOS, PFOA, PFHxS or PFNA in blood plasma not exceeding the guide value of 2.5 µg/L is actually related to meat, kidney and liver conformity for these 4 PFAS. It is therefore recommended to determine and to collect paired data (from the same animal) concerning PFAS in the blood plasma, muscle tissue, liver and kidney of bovines originating from businesses affected by a PFAS

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contamination. These additional data can then serve as a basis for confirming or, if necessary, adjusting the guide value of 2.5 µg/L.

If an analysis of a bovine blood plasma would show that the concentration of PFOS, PFOA, PFHxS or PFNA is higher than the guide value of 2.5 µg/L, then it is recommended to perform another blood plasma analysis at a later date until a result lower than this value is obtained. This later data can be calculated using the following equation:

$$withdrawal\ time\ t = \frac{(\log C_0 - \log 2{,}5) \times 2.303}{(0{,}693/DT_{50})}$$

Equation of the withdrawal time (t, in days) in blood plasma/serum to obtain the targeted guide value, where C_0 represents the starting concentration (µg/L) of PFOS, PFOA, PFHxS or PFNA in the blood plasma/serum, i.e. the concentration measured at initial sampling; 2,5 µg/L the suggested blood plasma/serum concentration guide value; DT_{50} the blood plasma/serum elimination half-life of PFOS, PFOA, PFHxS or PFNA (days).

Here, the Scientific Committee also refers to the uncertainties discussed above. The withdrawal time should therefore be considered as an indication to determine a time when a second blood test may be useful. However, it cannot be excluded that after the estimated withdrawal time a bovine would still show a PFAS level in blood plasma higher than 2.5 µg/L. In addition, calculated withdrawal times are expected to be rather long based on the relatively long elimination half-lives of PFAS described in the literature. For these reasons, the practical usefulness of these withdrawal times is rather low.

In addition, for PFOS, it is recommended to use the following ratios to estimate its relative concentration in bovine muscle tissue, liver or kidney without having to carry out separate measurements in all these tissues: the PFOS concentration in liver is on average 21 times (between 6 and 34 times) higher than that in muscle tissue, while the PFOS concentration in kidney is on average 8 times (between 3 and 12 times) higher than that in muscle tissue. Significant uncertainties are applicable to these ratios. The Scientific Committee recommends sampling paired samples (from the same animal) of muscle tissue, liver and kidney from Belgian bovines in the near future to verify these PFOS ratios between the different animal tissues.

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1. Terms of Reference

Questions

Three questions are asked to the Scientific Committee:

(1) From what levels of PFAS compounds (PFOS, PFOA, PFNA and PFHxS) in the blood of a bovine would a non-conformity (according to Regulation (EU) 2023/915) for the meat or offal (e.g. liver and kidneys) originating from that animal be suspected?
(2) What are the correlations between PFAS levels in muscle tissue, liver and kidneys of bovines?
(3) What are the depletion kinetics of PFAS (PFOS, PFOA, PFNA and PFHxS) in the blood of live bovines once the sources of PFAS contamination have been removed?

Legal provisions

Regulation (EC) No 1069/2009 of the European Parliament and of the Council of 21 October 2009 laying down health rules as regards animal by-products and derived products not intended for human consumption and repealing Regulation (EC) No 1774/2002 (Animal by-products Regulation)
Commission Regulation (EU) 2023/915 of 25 April 2023 on maximum levels for certain contaminants in food and repealing Regulation (EC) No 1881/2006

Method

The opinion is based on a literature study and on control data from the FASFC, on expert opinion and on regression analysis of a selected dataset.

2. Definitions and Abbreviations

PFAS	poly- and perfluoroalkyl substances
PFOS	perfluorooctanesulphonate
L-PFOS	linear PFOS
m-PFOS	mono-branched PFOS
di-PFOS	di-branched PFOS
t-PFOS	total PFOS
PFOA	perfluorooctanoic acid
PFNA	perfluorononanoic acid
PFHxS	perfluorohexanesulphonic acid
FVST	Danish Veterinary and Food Administration
DTU	Technical University of Denmark
PBPK	physiologically based pharmacokinetic
igPBPK	interactive generic PBPK
LOQ	limit of quantification
PFBS	perfluorobutanesulphonic acid
PFDA	perfluorodecanoic acid
PFHpS	perfluoroheptanesulphonic acid
FOVP	Front Office Food and Product Safety
BW	*body weight*
SciCom	Scientific Committee

Having regard to the discussions at the working group meeting of 08/10/2024 and the plenary sessions of the Scientific Committee of 29/11/2024 and 20/12/2024,

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the Scientific Committee provides the following opinion:

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3. Introduction

In 2023, sampled meat and liver from a few Belgian cows at the slaughterhouse did not meet the maximum levels for PFOS described in European legislation. These post-mortem results have led to a ban on the entry into the food chain of food obtained from animals from the affected businesses.

To track the elimination of PFOS, PFOA, PFNA and PFHxS during the life of a bovine, PFOS, PFOA, PFNA and PFHxS (which are the 4 PFAS for which maximum levels apply at European level in food of animal origin) can be measured in the blood of a live animal after the end of exposure to PFAS. There is no legal limit for these substances in the blood of cattle in Europe, but it could be proposed to set threshold values for each of the 4 congeners in the blood of cattle. These thresholds would allow predicting PFOS, PFOA, PFNA and PFHxS levels in meat and offal that are likely to be lower than the applicable legal maximum limits for these respective matrices.

The FASFC wants to propose tools to livestock farmers in order to be able to manage the contamination with PFAS as well as possible and to be able to make the best possible decisions in the context of their self-checking system.

After all, if a non-conformity with the European maximum levels for PFAS is found in meat or offal from bovines, the entire herd is placed under supervision by the FASFC. Currently, a compliant analysis result for PFAS in the meat and/or organs of slaughtered animals is mandatory so that animals from the herd can enter the food chain.

In this context, the answers to the three questions asked to the Scientific Committee would allow:

(i) for the farmer, to decide on the basis of a non-lethal blood test whether or not to send an animal to the slaughterhouse (with a defined probability that the meat and offal comply with the legal maximum levels for PFAS), and to set a time limit after a blood test that indicates a too high level of PFAS after which a second blood sample must be taken and analysed; and

(ii) for the authorities, whether it is necessary to destroy the liver or kidneys if the meat is non-compliant or vice versa, which may indirectly allow the operator to save on additional analyses.

4. Guide values of PFAS concentrations in bovine blood to predict compliant levels in meat or offal

This section first gives an introduction to the procedure used by the FASFC in the event of a non-conformity for PFAS in meat or offal from bovine animals. Then the current European maximum levels for PFAS in meat and offal from cattle are explained. It then discusses Danish legislation setting guide values for maximum PFOS concentrations in serum from bovines and sheep, which make it possible to predict legally permitted levels in meat and offal. Literature data and control data from the FASFC on PFAS levels in the blood, meat and offal of bovines are then discussed. Finally, based on a selection of data, a linear regression analysis was implemented to estimate, with a defined probability, a PFAS concentration value in bovine blood that is predictive of compliant levels in the meat or offal.

In the scientific studies found, PFAS levels in blood are sometimes measured in whole blood, but usually in plasma or serum. The FASFC control data relate to PFAS analyses of bovine blood serum. In order to be able to compare the data expressed by amount of plasma or serum, this opinion assumes that PFAS values in serum and plasma are equivalent. For the associated uncertainties, please refer to section 7.

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4.1. Procedure to be applied by the FASFC in the event of a non-conformity for PFAS in meat or offal of bovines

The FASFC applies the following step-by-step procedure in the event of a non-conformity for PFAS in meat or offal of bovines:

a) A bovine from an operator with no known PFAS contamination on his business:
   - a PFAS analysis of beef can be carried out in the slaughterhouse as part of the FASFC control plan;
   - in the meantime, the carcass will already enter the food chain;
   - if a non-conform result is found, the carcass or animal products derived from it will be traced, withdrawn from the market and, if necessary, recalled from the consumer (not always possible; may depend on mixtures of animal products and/or the measured value);
   - after which the carcass and/or animal products will be processed as Category 1 by-products; and
   - the bovines in the herd of the operator concerned are blocked.

b) Beef coming from an operator with a recently established PFAS contamination on his business (blocked herd):
   - a farmer may offer an animal to a slaughterhouse subject to the agreement of the FASFC;
   - a meat sample is taken for PFAS analysis in the slaughterhouse and the carcass is frozen pending the result; and
   - if the result is compliant, the carcass may be placed in the food chain.

c) When a blood test is taken and the measured PFAS concentration in blood is assessed in relation to the future guide value according to this draft opinion, procedure b) will still be followed because:
   - the time to obtain the results of the blood test may be long, with a risk of additional contamination in the meantime; and
   - slaughter cannot take place immediately after blood collection, during which time further contamination may occur.

Particular disclaimer: the above procedure applied by the FASFC may be altered on a case-by-case basis and on the basis of the outcome and recommendations of the present opinion. Furthermore, the manner in which a PFAS guide value will be used in official controls by the FASFC is still to be decided and can change in the course of time.

A guide value for PFAS in bovine blood would enable cattle farmers to manage their herd more effectively and select the animals they send to slaughter more accurately.

In terms of waste handling, carcasses and animal products from bovines that are non-compliant for PFAS are treated as hazardous waste (Category 1 animal by-products; Regulation (EC) No. 1069/2009, Art. 8d). This processing involves burning the products, with the possibility of converting them into products (bone meal and melted fat) for later energy production. In no case may animal by-products of Category 1 be used for the manufacture of animal feed or organic fertilisers. This way of handling waste reduces the risk of the PFAS compounds being released back into the environment.

4.2. Maximum levels of PFAS in bovine meat and offal according to Regulation (EU) 2023/915

For the meat of bovines (as well as of pigs and sheep), maximum levels of 0.30 µg/kg for PFOS, 0.80 µg/kg for PFOA, 0.20 µg/kg for PFNA, 0.20 µg/kg for PFHxS, and 1.3 µg/kg for the sum of PFOS, PFOA, PFNA and PFHxS apply according to Regulation (EU) 2023/915. For the offal of bovines (which includes liver and kidneys) as well as of pigs, sheep and poultry, maximum levels of 6.0 µg/kg for PFOS, 0.70 µg/kg for PFOA, 0.40 µg/kg for PFNA, 0.50 µg/kg for PFHxS, and 8.0 µg/kg for the sum of PFOS, PFOA,

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PFNA and PFHxS apply under the same Regulation. For details of these maximum levels, please refer to the most recent version of the Regulation.

4.3. Danish Decree No. 1386 of 29/11/2023: Guide values for a threshold concentration of PFOS in serum of bovine and ovine animals allowing to predict the conformity of meat

Danish Decree No. 1386 of 29/11/2023 gives guide values for a maximum PFOS concentration[1] in the blood serum of cattle (3.3 µg/L) and sheep (6.7 µg/L), which allow to predict a low risk of exceeding the legal maximum levels for PFOS in meat if the serum PFOS levels are below these guide values. However, this Danish decree does not give any guide values in blood (serum) for the other three PFAS. In addition, the guide values in the Danish decree apply specifically to bovine meat and not to offal.

The Scientific Committee received feedback from the Danish Veterinary and Food Administration (FVST) in December 2024 on the findings of enforcing the guide value of 3.3 µg PFOS/L in bovine blood serum. The FVST stated that this guide value has helped both itself and farmers to assess whether animal meat can contain PFOS levels that exceed the European maximum level under Regulation (EU) 2023/915. Furthermore, the FVST communicated that the guide value was set conservatively and that, to their knowledge, there was no exceedance of the European maximum level for meat when the guide value for blood was complied with. Animals with blood levels above the guide value may not be slaughtered in Denmark unless they have a special authorisation from the FVST. In a number of cases, PFOS levels were measured in the blood serum, meat and liver of the same animal. For one animal, a blood serum concentration of 1.4 µg PFOS/kg, a meat concentration of 0.12 µg PFOS/kg and a liver concentration of 4.4 µg PFOS/kg were measured. For another animal, a serum level of 1.4 µg PFOS/kg, a meat level of 0.05 µg PFOS/kg and a liver level of 4.0 µg PFOS/kg were found. In both cases, a blood level of PFOS below the guide value was therefore a good indication of compliance of the meat and liver with the European maximum levels.

4.4. Assessment by the Front Office Food and Product Safety (2024) from the Netherlands: Guide values for a threshold concentration of PFOS in the blood plasma of bovine animals and sheep allowing to predict conformity of meat and liver

On 14/12/2024, the Front Office Food and Product Safety (FOVP) from the Netherlands issued an assessment on the transfer and kinetics of PFOS in lamb, sheep and cattle (FOVP, 2024). The reason for this assessment was the notification to the Netherlands Food and Consumer Product Safety Authority by the FASFC that blood and meat from Belgian cattle, allegedly grazed in the 'Verdronken Land van Saeftinghe' in Zeeland (Netherlands), contained too high levels of PFOS.

4.4.1. Sheep

The FOVP predicted through a model that the plasma PFOS content of an ewe at the start of lactation may be up to 56 µg/L and 9.9 µg/L without exceeding the European maximum levels (Regulation (EU) 2023/915) for PFOS in meat and liver respectively in a 7-month-old lamb. The plasma PFOS content of an ewe at the start of lactation may be up to 17 µg/L and 3 µg/L without exceeding the maximum levels for PFOS in meat and liver of a 3-month-old lamb respectively. Furthermore, the model predicted that PFOS in lamb plasma should not exceed 6.6 µg/L or 1.1 µg/L to ensure that the maximum levels in meat and liver, respectively, are not exceeded. This is independent of the age of the lambs. In addition, the model predicted that PFOS in the plasma of an ewe should not exceed 6.6 µg/L or 1.1 µg/L to ensure that the maximum levels in the meat and liver respectively are not exceeded (FOVP, 2024).

[1] The guide values apply to the sum of linear and branched stereoisomers of PFOS, whether or not chromatographically separated.

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4.4.2. Cattle

For the determination of guide values in bovine blood plasma, the FOVP used relationships between PFOS levels in meat/liver and plasma reported in the literature. No transfer models were used. For bovine meat, the FOVP estimates that the European maximum level (Regulation (EU) 2023/915) for PFOS in bovine meat is exceeded at a plasma PFOS level above 5.0 µg/L. The maximum level for PFOS in the liver is estimated to be exceeded when the PFOS plasma level is higher than 5.5 µg/L. This is based on a median PFOS ratio for liver/plasma of 1.1 and for meat/plasma of 0.06, reported for cattle. Based on the highest reported meat/plasma (=0.08) and liver/plasma (=1.7) ratios according to the FOVP, the maximum PFOS levels in meat and liver are exceeded at plasma PFOS levels of 3.8 and 3.5 µg/L respectively. Based on the lowest reported meat/plasma (=0.01) and liver/plasma (=0.2) ratios according to the FOVP, the maximum levels for PFOS in bovine meat and liver are exceeded at plasma PFOS levels of 30 µg/L (FOVP, 2024).

4.5. Partition coefficients of PFAS in animal tissues with respect to blood on the basis of literature

4.5.1. Partition coefficients at the origin of the guide values proposed by Danish Decree No. 1386 of 29/11/2023

The FVST has based itself on two scientific reports from the Technical University of Denmark (DTU) from 2021 to determine the guide values proposed by Danish Decree No. 1386 of 29/11/2023.

A first DTU report from 2021 discusses the transfer of PFOS from water and grass to cattle and sheep (Vinggaard, 2021)[2], and gives partition coefficients of PFOS between different tissues (muscle, liver and kidney) and the blood in cattle and sheep (partition coefficient = concentration in tissue/concentration in blood plasma or blood serum). A second report from the DTU from 2021[3] discusses the transfer of different PFAS compounds from water and grass to cattle, and in addition to the partition coefficients for PFOS already discussed in the first report, also shows partition coefficients for PFOA and PFHxS.

The partition coefficients of the two DTU reports are based on different literature sources and taken from Table S2 of the review article by Death *et al.* (2021) (Appendix 4). The PFAS levels in blood used to determine these partition coefficients for cattle and sheep were, depending on the original scientific study (see Appendix 4), measured in plasma (Kowalczyk *et al.*, 2012; Kowalczyk *et al.*, 2013; Lupton *et al.*, 2014; *Lupton et al.*, 2015) or in whole blood (Vestergren *et al.*, 2013). The whole blood PFAS levels from *Vestergren et al.* (2013) were converted to serum concentrations by Death *et al.* (2021) (see Appendix 4). However, in the article by Death *et al.* (2021), the partition coefficients are all expressed as the ratio of PFAS in tissues to blood serum, while the DTU reports all express the same partition coefficients as the ratio of PFAS in tissue to blood plasma. These two approaches are probably based on the assumption from DTU's first report (Vinggaard, 2021) that concentration ratios of PFAS in serum and plasma can be considered 1:1.

The partition coefficients (concentration in tissue/concentration in blood serum) for cattle and sheep as described in the article by Death *et al.* (2021) are shown in Table 1.

[2] This report has been prepared in Danish. A machine translation into English is shown in Appendix 1.
[3] The original Danish version and the machine translation into English can be found in appendices 2-3.

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Table 1. Partition coefficients (ratio of PFAS concentrations between the tissues and the blood serum) for cattle and sheep as described in the article by Death *et al.* (2021)

	partition coefficient (conc. PFAS tissue/conc. PFAS blood serum)								
	PFOS			PFOA			PFHxS		
	muscles	liver	kidneys	muscles	liver	kidneys	muscles	liver	kidneys
sheep	0.15	5.1	1.1	0.02	0.2	0.4	/	/	/
cattle	0.02-0.09	0.3-1.6	0.07-0.6	0.07; 0.2	0.3; 1.2	1.0	0.05	0.16	0.3

For PFAS other than PFOS, PFOA and PFHxS, the second report of the DTU opted to follow a conservative approach and consequently to allocate to cattle the same partition coefficients as to PFOS, which, according to the DTU, is a worst case approach.

It can be inferred that the worst-case partition coefficient (concentration of PFOS in meat/concentration of PFOS in blood serum) of 0.09 (see Table 1) was applied to come from the legal maximum level in meat of 0.3 µg PFOS/kg (Regulation (EU) 2023/915) to a guide value of 3.3 µg PFOS per litre of blood serum proposed by the Danish Decree.[4]

4.5.2. Johnston *et al.* (2023)

Johnston *et al.* (2023) developed an approach to enable ante-mortem screening of dairy cattle suspected of being infected with elevated concentrations of PFOS in their muscle tissue. For this purpose, the PFOS concentration was determined in paired blood plasma and muscle tissue samples from 28 cows. The relationship between the concentration of PFOS in blood plasma and muscle tissue was found to depend on the phase of the cow (heifer, dairy cow or dry) and also on the time the exposure to PFOS was stopped.

Several agencies from the United States established an interim screening level of 4.1 µg/kg PFOS in raw beef. Johnston *et al.* (2023) determined that among the 28 cows studied, at least 95% of (i) heifers with a PFOS plasma concentration of 54 µg/L, of (ii) lactating cows with a PFOS plasma concentration of 30 µg/L, and of (iii) the dried cows with a PFOS plasma concentration of 18 µg/L would have a PFOS muscle concentration of less than 4.1 µg/kg.

The maximum level of 0.3 µg/kg for PFOS in beef according to Regulation (EU) 2023/915 is significantly lower than the limit value of 4.1 µg/kg used in the study by Johnston *et al.* (2023). Accordingly, the guide value for PFOS plasma concentration from the Danish Decree (3.3 µg/L) is significantly lower than the guide values stated in Johnston *et al.* (2023). The raw data from the study by Johnston *et al.* (2023), accessible in the supporting information of the publication, were used by the SciCom to calculate partition coefficients of PFOS in the meat and blood plasma of each animal. These range from 0.03 to 0.11.[5] These values are in the same order of magnitude as those discussed in the DTU reports from 2021.

4.6. Assessment of FASFC control data

4.6.1. Presentation of FASFC control data

In 2023, the FASFC identified non-conformities of PFOS in beef from cows at three different Belgian operators. For each of the samples, the levels of the four PFASs of interest were measured, namely PFOS, PFOA, PFNA and PFHxS. At a first operator, a non-compliant PFOS level of 0.53 µg/kg was found in the meat of a 4-year old dairy cow. At a second operator, a non-conform level of 5 µg/kg PFOS was

[4] It is assumed that the ratio between PFAS levels in blood serum and plasma is 1:1.

[5] All partition coefficients calculated based on data from Johnston *et al.* (2023) are shown in Appendix 5.

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found in a meat sample from a 7-year-old dairy cow, which also resulted in a non-conformity with the legal maximum level of the sum of the four PFAS. A third operator measured a non-compliant PFOS level of 1.2 µg/kg in a meat sample from a 3.5-year-old beef cow. None of the meat samples from the three cows were found to exceed the legal maximum levels of PFOA, PFNA and PFHxS.

Subsequently, the blood serum[6] (ante-mortem), the meat and the liver (post-mortem) of six dairy cows and a bull were sampled from two of the three operators. Details of the results of PFAS analyses in meat, liver and blood from cattle can be found in Appendix 6.

For PFOA, PFNA and PFHxS, no exceedances of the legal maximum levels were observed for the meat and liver samples. Relatively low serum concentrations of these PFAS were found. Except for one dairy cow, all meat and liver samples were found to be non-compliant for PFOS. The non-conform samples had PFOS levels of 0.56 to 3.9 µg/kg in meat and 19 to 93 µg/kg in liver. Although non-compliant, it is striking that the PFOS levels in the meat and liver of the bull are lower than for all non-compliant dairy cows. Except for the meat sample of the bull, the non-conform results of PFOS in meat and liver always lead to non-conform results for the sum of the 4 PFAS.

4.6.2. Verification of partition coefficients calculated on the basis of FASFC control data against those mentioned in the literature

Table 2 shows the partition coefficients (PFAS concentration in tissue/PFAS concentration in serum) of different PFAS between meat and blood serum of bovine animals as calculated from the FASFC control results. These partition coefficients are very similar to those from the literature. The highest partition coefficient based on the FASFC results is rounded to 0.09, which is identical to the value from the literature selected by the FVST in Denmark, and is very close to the highest partition coefficient of 0.11 that could be derived from the study by Johnston *et al.* (2023).

The partition coefficients of PFAS between the liver and serum based on the FASFC results are similar to those from the literature, except for the relatively higher value of 3.115 obtained for the bull.

For PFOA, PFNA and PFHxS, FASFC results were often below the detection limit; the calculations were therefore carried out with the detection limits as the concentration (using an upper bound approach), which is a source of uncertainty. The partition coefficients obtained for PFHxS also vary greatly.

[6] No maximum limits are available for blood (live collection).

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Table 2: Partition coefficients of PFAS between bovine tissues and blood serum, as calculated based on FASFC analysis results obtained for samples taken from cattle from two operators during the period December 2023-May 2024. For further details of the samples, please refer to Appendix 6.

	partition coefficients of PFAS between the meat (m) and the blood serum (s)			
cattle identification	PFOSm/PFOSs	PFOAm/PFOAs[e]	PFNAm/PFNAs[e]	PFHxSm/PFHxSs[e]
bovine 1 [a,c]	0.048	0.500	0.500	0.098
bovine 2 [a,c]	0.041	0.500	0.500	0.128
bovine 3 [a,c]	0.075	0.500	0.250	0.357
bovine 4 [b,d]	**0.092**	0.500	0.250	0.500
bovine 5 [a,c]	0.038	0.500	0.250	0.091
bovine 6 [a,c]	0.040	0.500	0.250	0.143
bovine 7 [a,c]	0.050	0.500	0.250	0.077
	partition coefficients of PFAS between the liver (l) and the blood serum (s)			
cattle identification	PFOSl/PFOSs	PFOAl/PFOAs	PFNAl/PFNAs	PFHxSl/PFHxSs[e]
bovine 1 [a,c]	0.873	1.000	1.000	0.196
bovine 2 [a,c]	1.019	1.000	1.000	0.256
bovine 3 [a,c]	**1.700**	1.000	0.500	0.714
bovine 4 [b,d]	**3.115**	1.000	0.500	1.000
bovine 5 [a,c]	1.103	1.000	0.500	0.182
bovine 6 [a,c]	1.069	1.000	0.500	0.286
bovine 7 [a,c]	1.077	1.000	0.500	0.154

[a] bovine animals from the first operator; [b] bovine animals from the second operator; [c] dairy cow; [d] bull; [e] for individual analysis results, the PFAS concentrations below the detection limit were equated to the relevant detection limit for the calculations of the partition coefficients (upper bound)

4.6.3. Evaluation of the guide value for PFOS concentration in blood serum proposed by Danish Decree No. 1386 of 29/11/2023 against the FASFC literature and control data

Sections 4.5.2 and 4.6.2 show that the meat/blood PFOS partition coefficient (muscle PFOS concentration/blood PFOS concentration) on which the guide value of 3.3 µg PFOS/L blood plasma as applied by the Danish authorities is based is confirmed by the findings in Johnston *et al.* (2023) and the FASFC control data.

Similarly, guide values for other PFAS in blood plasma could also be derived based on the legal European maximum levels and the partition coefficients between meat and blood of cattle published in the literature. In addition to meat, a guide value could also be derived to be able to assess the conformity of liver or kidneys.

Table 3 shows a selection of literature partition coefficients for PFOS, PFOA and PFHxS in meat, liver and kidneys compared to in blood serum. For each combination of PFAS and animal tissue, the worst case, i.e. the highest, partition coefficients from the article by Death et al. (2021)[7] were selected. Furthermore, the table shows the target values calculated by the Scientific Committee for a maximum

[7] However, an exception was made for the partition coefficient of PFOS in beef compared to blood plasma. Indeed, from the results of Johnston *et al.* (2023), an even higher partition coefficient of 0.11 could be derived for the ratio of PFOS in muscle tissue to blood plasma, than the value of 0.09 on which the Danish authorities' guide value is based. As these values are close together, it is nevertheless chosen to give preference to the value used to draw up the Danish decree.

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serum concentration of PFOS, PFOA and PFHxS. These guide values were calculated by dividing the maximum levels of PFAS in muscle, liver and kidneys as described in Regulation (EU) 2023/915 by the selected partition coefficients.

Table 3: Selection of partition coefficients (PC) of PFAS between animal tissues and bovine blood serum, using the highest coefficients found in Death *et al.* (2021), and proposal of guide values for a maximum PFAS concentration in blood serum, calculated based on the maximum levels of PFAS in the relevant animal tissues according to Regulation (EU) 2023/915

	PFOS			PFOA			PFHxS		
	muscles	liver	kidneys	muscles	liver	kidneys	muscles	liver	kidneys
maximum levels (EU) 2023/915 (µg/kg)	0.30	6.0	6.0	0.80	0.70	0.70	0.20	0.50	0.50
PC (conc. PFAS tissue/conc. PFAS blood serum)	0.09 [a,b]	1.6[b]	0.6[b]	0.2[b]	1.2[b]	1.0[b]	0.05[b]	0.16[b]	0.3[b]
guide value blood serum (µg/L)[c]	3.3	3.8	10	4.0	0.6	0.7	4.0	3.1	1.7

[a] An even higher partition coefficient of 0.11 can be derived from the results of Johnston *et al.* (2023). Since 0.09 is the value on which the Danish authorities' guide value is based (and this is also close to the value of 0.11), it is nevertheless chosen to give preference to the value used to draw up the Danish decree.

[b] source = Death *et al.* (2021)

[c] The guide values in blood serum are determined by dividing the maximum concentration (µg/kg) of a given PFAS in a tissue according to Regulation (EU) 2023/915 by the partition coefficient (PC: concentration of PFAS in a tissue/concentration of PFAS in blood serum).

In the assessment of the FOVP of 10 December 2024 (FOVP, 2024), reference was made to an unpublished study by Wageningen Food Safety Research carried out by Hoogenboom *et al.* (2024). In this study, for dairy cows (n=2) experimentally exposed to PFOS, a meat-plasma partition coefficient of PFOS of 0.05 and a liver-plasma partition coefficient of PFOS of 1.4 were observed. In the same study, 84 days after discontinuation of exposure to PFOS, the dairy cows (n=2) showed virtually unchanged coefficients (0.05 between meat and plasma, and 1.2 between liver and plasma). These results are in line with the partition coefficients shown in Table 3.

The guide value for a PFOS content in bovine serum (3.3 µg/L) for the evaluation of conformity in meat proposed by the Danish authorities (section 4.3) is more conservative than the values proposed by the FOVP, i.e. 3.8 µg/L and 3.5 µg/L blood plasma, for the evaluation of conformity in meat and liver respectively (section 4.4.2) (assuming that PFOS values in serum and plasma are equivalent).

The guide value in blood serum for estimating a compliant PFOS level in muscle is lower than that for the liver and kidneys (Table 3). The guide value of 3.3 µg PFOS/L blood plasma applied by the Danish authorities is therefore a good starting point for assessing conformity in the meat or offal (liver, kidneys) of a cattle. For PFOA and PFHxS, less literature data is available on partition coefficients (see also Death *et al.*, 2021). Therefore, guide values calculated for this purpose in blood serum are less reliable.

For PFNA, no partition coefficients are known from the literature. The analyses of the FASFC also showed that PFNA was not detected in the samples, which means that they cannot be used for an accurate estimate of partition coefficients. For PFNA, according to the conservative approach of the DTU (2021), the partition coefficient of PFOS can be used for muscle tissue. This could also be done for liver. Consequently, the guide values for PFOS (for muscle tissue, liver and kidneys) could also be used for PFNA.

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4.7. Linear regression analysis to estimate the relationship between a compliant PFAS level in the meat or offal of bovines based on a PFAS concentration in bovine blood

In this section, a linear regression analysis was carried out to predict a PFAS level in bovine meat according to the legal maximum levels based on a PFAS concentration in bovine blood. The logarithm of PFOS levels in blood plasma and muscle tissue from the same animals (paired results) from the study by Johnston *et al.* (2023) were chosen as raw data for the calculations (see Appendix 5). These paired data were obtained from 28 animals, including 10 heifers, 13 lactating cows and 5 dry cows.

As indicated above (section 4.5.2), Johnston *et al.* (2023) base themselves on a preliminary screening level of 4.1 µg PFOS/kg meat while the legal European maximum level (Regulation (EU) 2023/915) for PFOS in meat is 0.3 µg/kg.

A linear regression analysis was applied to the data after a logarithmic transformation. No distinction was made for the lactation status of the cows due to the limited sample size (n=28). The details of the linear regression analysis are shown in Appendix 7.

By applying the regression parameters, it can be determined that a PFOS content of 0.3 µg/kg meat (= legal European maximum content) corresponds to a most likely PFOS content of 6.2 µg/L blood plasma, with a 95% confidence interval of 2.5 to 23.4 µg/L. The very wide confidence interval reflects the uncertainty associated with the value of 6.2 µg/L and is explained by the fact that the concentration values are generally between 1 and 10 µg/kg PFOS in muscle and are therefore higher than the European maximum level of 0.3 µg/kg. As a precaution, the Scientific Committee recommends using the lower limit of this confidence interval, namely 2.5 µg/L blood plasma, as a guide value.

For the liver and kidneys, it was not possible to apply this type of approach due to the lack of paired data (PFAS levels in the blood and tissues of the same animal).

4.8. SciCom proposal for a guide value for PFAS in blood for estimating a compliant PFAS content in meat or offal from cattle

This lower limit of 2.5 µg PFOS/L blood plasma is close to and at the same time more conservative than the guide value of 3.3 µg PFOS/L blood serum as applied by the Danish authorities and which is verified by literature and FASFC control data. The Scientific Committee therefore considers that a PFOS level of 2.5 µg/L in blood plasma is an appropriate guide value for assessing conformity of PFOS levels in meat, liver and kidneys with the legal European maximum levels.

As explained above in section 4.6.3, according to the conservative approach of the DTU (2021), the partition coefficient of PFOS, in the absence of data for PFNA, can also be used for the latter. The Scientific Committee therefore considers that 2.5 µg/L blood plasma is also an appropriate guide value for PFNA.

Table 3 above gives possible guide values for PFOA of 4.0 µg/L, 0.6 µg/L and 0.7 µg/L blood serum to estimate conformity in muscle, liver and kidneys, respectively. The value of 2.5 µg/L blood plasma for PFOS is higher than some of these values. However, the values for PFOA are based on a (very) limited number of literature data (Appendix 4).

The SciCom therefore also recommends the general guide value of 2.5 µg/L blood plasma for PFOA. Finally, it is noted that the same reasoning is valid for PFHxS.

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5. Relationships between PFAS levels in bovine muscle tissue, liver and kidneys

The relationships between PFAS levels in bovine muscle tissue, liver and kidneys are assessed in this section using (i) FASFC control data (Appendix 6) and (ii) data from the scientific literature.

Few data are available in the literature on the concentrations of the 4 PFAS in the muscle tissue, liver and kidneys of the same animal. Most data relate to PFOS. The data collected and the results of the calculations of the liver/muscle tissue and kidney/muscle tissue concentration ratios are shown in Appendix 7.

The FASFC provided data for 7 animals, including 6 dairy cows (5-8 years old) and 1 bull (2.5 years old), for the 4 PFAS in muscle tissue and liver, but not in the kidneys. The results show that only PFOS was detectable in muscle tissue and liver and therefore no detection of the other 3 PFAS compounds (PFOA, PFHxS and PFNA). Liver-to-muscle concentration ratios for PFOS range from 18 to 29 for dairy cows and is 34 for bulls.

In Kowalczyk *et al*. (2013) (Germany), 6 dairy cows (average weight of 583 kg) were fed PFAS-containing feed for 28 days. Three cows were slaughtered after these 28 days (group 1) and the three remaining cows received 21 days of PFAS-free feed before being slaughtered (group 2). Concentrations of PFHxS, PFOS, PFBS and PFOA were determined in muscle tissue *(longissimus dorsi)*, liver and kidneys of the 6 animals. The available data for these tissues are averages and standard deviations for the 3 animals in each group. Based on these data, PFOS L/M ratios (concentration in liver to muscle tissue) of 20 and 22 were calculated for the 2 groups of dairy cows, while the K/M ratios (concentration in kidneys to muscle tissue) were 7 and 8, respectively.

Vestergren *et al.* (2013) conducted a study on a dairy farm over 2 years old in Sweden, in an environment representative of the background contamination with PFAS (no specific source of PFAS emissions in the environment). Concentrations of the 4 PFAS of importance in muscle tissue and liver were obtained for 5 dairy cows (between 29 and 45 months old, average weight 587 kg) which were sampled at slaughter. For PFOS, the ratio between the concentration in liver and muscle tissue was 6.

Lupton et al. conducted a study in the United States on 3 bulls (average weight 329 kg) which were administered PFOS *(Lupton et al.*, 2014) and ^{14}C-labelled PFOA (Lupton *et al.*, 2012). PFOA was not found in animal tissues (Lupton *et al.*, 2012). For PFOS, the concentration ratio L/M was equal to 16, and the K/M ratio was equal to 3 (Lupton *et al.*, 2014).

Lupton *et al.* (2015) conducted a study in the United States on 2 bulls (average weight 307 kg) and 4 heifers (average weight 308 kg). The two bulls and 4 heifers received a single dose of PFOS. Two heifers were slaughtered after 105 days (corresponding approximately to the half-life of PFOS in blood of 114 days as calculated in Lupton *et al.* (2014)). The remaining animals were slaughtered after 343 days (approximately 3 times the half-life). PFOS was measured in blood plasma and various tissues. For muscle tissue, the values of tenderloin are shown in Appendix 8. According to the data from this study, the concentration ratios of PFOS between liver and muscle tissue range from 16 to 24 and are between 8 and 12 for the concentration ratios between kidneys and muscle tissue.

In the publication by Drew *et al.* (2021), figures for the PFOS concentration in tissues are not available. For this opinion, however, they were derived from Figure 5 of the said publication in order to be able to calculate concentration ratios of PFAS between different tissues. The concentration ratio of PFOS in liver to muscle tissue is 7 for the 12 animals slaughtered 63 days after the end of exposure to PFAS,

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and 22 for the 5 animals slaughtered 215 days after the end of exposure. The ratios between the kidney and muscle tissues are 6 and 7, respectively.

Table 4 summarises the L/M and K/M concentration ratios calculated for PFOS. In this respect, for the three other PFAS for which European maximum levels apply (PFOA, PFNA and PFHxS), such calculations could not be carried out because either: (i) the data are insufficient, (ii) because they were not measured in the available studies, (iii) because they were not detected or were only present in very low concentrations, or (iv) because the available data comes from only one single study.

Table 4 shows that the PFOS concentration in the liver (n=16) is on average 21 times higher than that in the muscle tissue, while the PFOS concentration in the kidneys (n=8) is on average 8 times higher than that in the muscle tissue.

Relatively large standard deviations are associated with these average ratios. This variation is not surprising, as we are talking about animals from different farms, different animals (dairy cows, beef cattle, different ages) that have been exposed to different amounts of PFAS through different sources. In short, these are field data, not the result of controlled experiments. The study in the Netherlands also reports this variation (FOVP, 2024). Possible explanations for this are also given in the chapter on uncertainties (section 7).

To apply regression analysis to these PFOS data, there are in principle too few data available. As an illustration, this was nevertheless attempted and the results are shown in Appendix 9.

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Table 4: PFOS concentrations (µg/kg) in bovine tissues and concentration ratios of PFOS between liver (L) and muscle tissue (M) and between kidneys (K) and muscle tissue (M)

		PFOS					
Animals	**Age or weight**	**M (µg/kg)**	**L (µg/kg)**	**L/M**	**K (µg/kg)**	**K/M**	**Source**
dairy cow	5 years old	3	55	18			Appendix 6
dairy cow	6 years old	2.2	55	25			Appendix 6
dairy cow	7 years old	0.3	6.8	23			Appendix 6
dairy cow	8 years old	3	86	29			Appendix 6
dairy cow	8 years old	3.5	93	27			Appendix 6
dairy cow	6.5 years old	3.9	84	22			Appendix 6
bull	2.5 years old	0.56	19	34			Appendix 6
dairy cow (n=3), group 1	583 kg	145	2952	20	1074	7	Kowalczyk *et al.* (2013)
STD		36	718		153		
dairy cow (n=3), group 2	583 kg	178	3964	22	1408	8	Kowalczyk *et al.* (2013)
STD		48	1035		303		
dairy cow (n=5)	29 to 45 months	0.021	0.13	6			Vestergren *et al.* (2013)
STD		0.02	0.032				
bull (n=3)	329 kg	1100	17900	16	3700	3	Lupton *et al.* (2014)
STD		100	2300		500		
bull (n=2) 343 days[b]	307 kg	7	149	21	82	12	Lupton *et al.* (2015)
STD		1	15		24		
dairy cow (n=2) 105 days[b]	308 kg	360	8760	24	3960	11	Lupton *et al.* (2015)
STD		40	3540		160		
dairy cow (n=2) 343 days[b]	308 kg	300	4740	16	2360	8	Lupton *et al.* (2015)
STD		10	560		570		
bovine animals *(n=12)* 63 days[b]	1.4-12.3 years	18	120	7	100	6	Drew *et al.* (2021)
heifers (n=5) 215 days[b]	1.4 - 2 years	4.5	100	22	32	7	Drew *et al.* (2021)
		mean		21		8	
		STD		7		3	
		minimum		6		3	
		median		22		8	
		maximum		34		12	
		n		16		8	

[a] Values derived from Figure 1 above; [b] wash-out period; M = Muscle tissue; L = liver; K = kidney; L/M = ratio between the concentration of PFOS in the liver and its concentration in muscle tissue; K/M = ratio between the concentration of PFOS in kidneys and its concentration in muscle tissue; n = number of results; STD = standard deviation

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6. Determination of a withdrawal time after which a second blood sample should be taken and analysed, in the event of an initial blood test that indicates a too high level of PFAS

For the operators, it would be an aid to be able to estimate a withdrawal time after an initial blood test that indicates a too high level of PFAS (compared to the target value of 2.5 µg/L), after which a second blood sample should be taken and analysed (and retested against the target value of 2.5 µg/L). Therefore, a literature study was carried out on the depletion kinetics of PFAS in bovine blood. Below the approaches found via (i) elimination half-lives and (ii) depletion models are discussed. Finally, a pragmatic approach was developed to calculate such a withdrawal time.

6.1. PFOS results (blood serum) from cattle grazed in the Verdronken Land van Saeftinghe (Netherlands)

The recent FOVP report (2024) reports on measurements carried out by the FASFC on cattle from a Belgian business that had animals in the Verdronken Land van Saeftinghe (Netherlands). The samples concerned were taken at the end of August 2024 and show PFOS levels in 7 animals from 42 to 86 µg/L serum (average 72 µg/L). PFOS levels of 22 and 24 µg/L were found in two bovines in the Verdronken Land van Saeftinghe in 2023, but not in 2024. This indicates a 68% reduction in 1 year by moving to a non-PFAS contaminated area. No further information is available on these cattle.

6.2. Depletion kinetics of PFAS in bovine blood: elimination half-lives

Several researchers have determined elimination half-lives for PFAS in bovine blood.

The review article by Death *et al.* (2021) reports elimination half-lives in blood of:

- 38.7 days for the elimination of PFOS in a lactating cow (Vestergren *et al.*, 2013);
- 120 ± 4.1 days (n = 4) for elimination of PFOS in a steer (Lupton *et al.*, 2015);
- 1.3 days for PFOA in a lactating cow (Vestergren *et al.*, 2013); and
- 19.2 ± 3.3 days (n = 4) for PFOA in bovines (Lupton *et al.*, 2012).

In addition, Vestergren *et al.* (2013) describes an elimination half-life of 8.7 days for PFNA in the blood of a lactating cow.

In Drew *et al.* (2021), serum elimination half-lives for different PFAS were determined for 16 cattle (beef cattle). The mean serum elimination half-lives are as follows:

- 74.1 ± 13.4 days for PFOS;
- 45.7 ± 9 days for PFHpS;
- 9.3 ± 1.3 days for PFHxS;
- 12.3 ± 3.2 days for PFNA; and
- 60.4 ± 10.4 days for PFDA.

Finally, *Lupton et al.* (2015) describes an elimination half-life for PFOS of 106 ± 23.1 days (n=4) in heifer plasma and van Asselt *et al.* (2013) an estimated elimination half-life for PFOS of 56 days.

It is striking that the elimination half-lives for PFOS in bovine blood are longer than those for other PFAS compounds. Furthermore, longer half-lives are observed in males compared to females. Primarily from the data for PFOS it can be inferred that milk release appears to promote the elimination of PFOS (decreasing elimination half-lives from bull to heifer to lactating cow). For further details of the studies in which the elimination half-lives of PFAS compounds in bovine blood were determined, such as the

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sampled matrix (whole blood, plasma or serum), the breed of the animals or the degree of exposure of the animals to PFAS, please refer to the table in Appendix 10.

6.3. Depletion kinetics of PFAS in bovine blood: depletion models

6.3.1. Chou *et al.* (2023)

Chou *et al.* (2023) describe a physiologically based pharmacokinetic (PBPK) model to predict tissue and milk residues and estimate withdrawal times for multiple PFAS including PFOA, PFOS and PFHxS in beef cattle and lactating dairy cows. Modelled predictions were usually within a factor of two of experimental data for blood plasma, tissues and milk with an estimated coefficient of determination (R^2) greater than 0.95.

Chou *et al.* (2023) converted the PBPK model into a web-based interactive generic PBPK (igPBPK) platform with a user-friendly dashboard where tissue and milk withdrawal interval estimates for PFOA, PFOS and PFHxS can be performed for beef and dairy cattle. The article by Chou *et al.* (2023) describes the platform as follows: "It is a web-based PBPK interface of the population PBPK model that can be used to estimate withdrawal intervals of PFOA, PFOS, PFHxS in beef cattle and dairy cows. This web-based interface has been designed with RShiny and is published online via the following link: https://pbpk.shinyapps.io/igPFAS_PBPK/. Users can enter the parameter values into the interface and run a Monte Carlo simulation with 1000 iterations to calculate PFOA, PFOS and PFHxS concentrations in plasma, muscle, liver, kidneys, fat and milk based on different exposure scenarios in beef and dairy cattle.[8] This platform could be used to predict withdrawal times for PFAS in beef and dairy cattle. In particular, the platform provides estimates of withdrawal times for muscle tissue, liver, kidneys and milk based on reported exposure to PFAS via contaminated soil and/or water. One limitation is that different parameter values must be known to be entered in the application, in particular the number of animals, the PFAS concentration in the soil or in the water, the time interval for dosing the PFAS contamination, the frequency of exposure and the duration of exposure. For a livestock farmer, it will often be difficult, if not impossible, to know or determine these values. In addition, the source of contamination may also be from a source other than soil or water, such as contaminated feed. In addition, the PFAS concentrations are assessed against the following estimated threshold values (which are higher than the maximum levels according to European legislation shown in section 4.2): 78.4 ng/g for liver, 10 ng/g for muscle tissue and 1719 ng/g for kidney in case of PFOS or PFOA, and 15.7 ng/g for liver, 2 ng/g for muscle tissue and 343 ng/g for kidney in case of PFHxS.

6.3.2. Johnston *et al.* (2023)

Johnston *et al.* (2023) developed a plasma depletion model to evaluate the estimated withdrawal time required to reduce PFOS in dairy cattle muscle tissue below a certain level of concern. The model development was based on plasma samples from dairy cattle exposed to PFOS. In particular, PFOS concentrations were obtained from 92 plasma samples from 8 cows, including 5 heifers and 3 lactating cows. The samples were taken approximately every two weeks from 0 to 153 days after the cows were moved to a non-contaminated environment after staying in the PFAS-contaminated environment (contaminated water and silage). A log-cubic model seemed to best fit the experimental data:

[8] However, it appears that no concentrations are calculated for fat on the platform.

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$$\ln(C_{it}) = b_0 + b_i cow_i + b_1 days + b_2 days^2 + b_3 days^3 \quad (1)$$

Equation 1. Log-cubic plasma depletion model according to Johnston *et al.* (2023) to evaluate the withdrawal time (in days) required to reduce PFOS in dairy muscle tissue below a certain level of concern, with C_{it} the plasma concentration of PFOS in cow i at time t, and cow_i a class variable for animal identification (so that each cow has an estimated intercept value at time zero (C_{i0})). It should be noted that the intercept terms ($b_0 + b_i\ cow_i$) represent the initial PFOS concentration before removal from exposure and are species specific. The non-intercept parameters (b_1, b_2 and b_3) determine the predicted behaviour of the depletion slope over time (positive terms of higher order (b_2 and/or b_3) result in convex (U-shaped) curves).

Such a model seems quite complex to apply in practice. In addition, the initial PFOS concentration before removal from the exposure zone must be known.

6.3.3. Mikkonen *et al.* (2023)

Mikkonen *et al.* (2023) have derived an equation for the depletion of PFAS compounds in bovine blood serum. For reasons of simplicity and ease of use, a first-order depletion kinetic was chosen with the following equation:

$$d\big(C_{serum}(t)Vd(t)\big)/dt = D_{daily} - kE(t) \times C_{serum}(t)Vd(t) \quad (2)$$

Equation 2. First-order equation for the depletion of PFAS in bovine serum, according to Mikkonen *et al.* (2023), with C_{serum} as the serum PFAS concentrations, Vd as the (apparent) volume of distribution, D_{daily} as the daily dose taken (µg) and kE as the first-order elimination rate constant (day $^{-1}$). For the (apparent) volume of distribution (Vd), values of 0.085 litres/kg body weight (BW) for PFOS and 0.14 litres/kg BW for PFHxS were selected by Mikkonen *et al.* (2023).

For the first-order elimination rate constant, the following equation is given:

$$kE(t)(1/d) = \ln(2)/\left[DT_{50} \times \left(\big(BW(t)\big)/MW\right)^{0,25}\right] \quad (3)$$

Equation 3. Equation for the first-order elimination rate constant (kE) of PFAS in bovine serum, according to Mikkonen *et al.* (2023), with DT_{50} as the serum half-life, BW as the body weight and MW as the mature weight.

In the study by Mikkonen *et al.* (2023), equation 2 was applied to PFOS and PFHxS, but not to PFOA due to its short half-life. As serum half-lives (DT_{50}), these of Drew *et al.* (2021) were selected, namely 74.1 days (mean value) for PFOS (non-lactating cattle) and 9.3 days (mean value) for PFHxS. For lactating cattle, the half-life of 56 days was selected according to van Asselt *et al.* (2013). Mikkonen *et al.* (2023) mention that for PFOS the elimination half-lives were not aggregated for meat and dairy cattle because Mogensen *et al.* (2015) showed that lactation increases the elimination for PFOS but not for PFHxS.

Using the assumptions of values for DT_{50} and for the (apparent) volume of distribution (Vd), equation 2 allows to estimate the approximate time that the level of PFOS or PFHxS falls below a certain value after removal of the contamination source. However, this approach is not suitable for PFOA and PFNA, as no values for V_d are known for these PFAS.

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6.3.4. Pragmatic approach to calculate a withdrawal time after which a second blood sample should be taken and analysed, in the event of an initial blood test that indicates a too high PFAS level

The models and equations discussed above in 6.3.1 to 6.3.3 appear to be little usable in practice due to the required parameters that must be known and the complexity. This section therefore proposes a pragmatic approach developed by the SciCom. This pragmatic approach has resulted in an equation which can be used to estimate a withdrawal time before a second blood sample should be taken and analysed, in the event of an initial blood test that indicates a too high level of PFAS. Below is a step-by-step explanation of how this equation was derived.

During the depletion phase of a xenobiotic substance, the relationship between blood plasma/serum concentration and time can be given by the following equation, based on first-order depletion kinetics:

$$C = C_0 \times e^{-k_e \times t} \qquad (4)$$

Equation 4. Equation of the blood plasma/serum concentration - time curve at first order depletion kinetics, where C_0 represents the starting concentration (µg/L) of PFOS, PFOA, PFHxS or PFNA in the blood plasma/serum, i.e. concentration measured at first sampling; k_e the elimination rate constant of PFOS, PFOA, PFHxS or PFNA (day $^{-1}$); C the target blood plasma concentration threshold value, i.e. the proposed guide value of 2.5 µg/L; and t the withdrawal time required to reach the target threshold (days).

When equation 4 is proposed in the form of natural logarithms, and then converted to decimal logarithms via ln C = 2.303 x log C, the following equation is obtained:

$$\log C = \log C_0 - \frac{k_e \times t}{2{,}303} \qquad (5)$$

Equation 5. Equation of the decimal logarithm of the blood plasma/serum concentration - time curve at first order depletion kinetics, where C_0 represents the starting concentration (µg/L) of PFOS, PFOA, PFHxS or PFNA in the blood plasma, i.e. concentration measured at first sampling; ke the elimination rate constant of PFOS, PFOA, PFHxS or PFNA (day^{-1}); C the target blood plasma concentration threshold value, i.e. the proposed guide value of 2.5 µg/L; and t the withdrawal time required to reach the target threshold (days).

When in equation 4, t is equated to DT_{50}, equation 4 becomes the following equation:

$$C_0/2 = C_0 \times e^{-k_e \times DT_{50}} \qquad (6)$$

Wherein solving this equation 6 and presenting it in the form of decimal logarithms yields the following equation:

$$k_e = 0{,}693/DT_{50} \qquad (7)$$

Finally, inserting equation 7 into equation 5 yields the following practical equation to calculate the withdrawal time (t) for any blood plasma/serum concentration to reach the proposed guide value of 2.5 µg/L, subject to applying the corresponding blood plasma/serum elimination half-lives for PFOS, PFOA, PFHxS or PFNA for bovine animals:

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$$\log C = \log C_0 - \frac{(0{,}693/DT_{50}) \times t}{2{,}303} \qquad (8)$$

or:

$$withdrawal\ time\ t = \frac{(\log C_0 - \log 2{,}5) \times 2{,}303}{(0{,}693/DT_{50})} \qquad (9)$$

Equation 9. Equation of the withdrawal time (t, in days) in blood plasma/serum to obtain the targeted guide value, where C_0 represents the starting concentration (µg/L) of PFOS, PFOA, PFHxS or PFNA in the blood plasma/serum, i.e. the concentration measured at the first sampling; 2.5 µg/L the proposed blood plasma/serum concentration guide value; DT_{50} the plasma/serum elimination half-life of PFOS, PFOA, PFHxS or PFNA (days).

Table 5 shows some calculation examples for PFOS, applied to the FASFC control data from Appendix 6.

Table 5: Calculation examples of the withdrawal time to reach the targeted guide blood plasma/serum concentration of 2.5 µg/L, applied to the PFOS data from Appendix 6.

Animals	Age	PFOS conc. blood serum (µg/L)	Plasma/serum elimination half-life DT_{50} (days)	Calculated withdrawal time according to equation 9, to reach the proposed targeted guide value of 2.5 µg/L (days)
dairy cow	5 years old	63	56 (van Asselt *et al.,* 2013)	260.8
dairy cow	6 years old	54	56 (van Asselt *et al.,* 2013)	248.3
dairy cow	8 years old	78	56 (van Asselt *et al.,* 2013)	278.1
dairy cow	8 years old	87	56 (van Asselt *et al.,* 2013)	286.9
dairy cow	6.5 years old	78	56 (van Asselt *et al.,* 2013)	278.1
bull	2.5 years old	6.1	74.1 (Drew *et al.*, 2021)	95.4

7. Uncertainties

This opinion is subject to the following sources of uncertainty:

- The consulted PFAS levels in blood, meat and offal from cattle are influenced by many factors, including:
  - the concentration of exposure (background exposure versus exposure to particularly contaminated hotspots);
  - duration of exposure (single dose, short-term or long-term exposure, or steady-state body load);
  - route/matrix of oral administration (food, water or bolus);
  - exposure to a single or a mixture of several PFAS compounds;
  - breed, sex, age and stage of growth of the animal;
  - production type and lactation status of bovine animals: dairy cattle (dried, lactating), beef cattle;
  - maternal transfer (40% for PFOS; 70% for PFHxS) (Mikkonen *et al.*, 2023);
  - the measurement uncertainty of the analytical methods used for analyses in biological tissues and liquids;
- the elimination of PFAS via milk and cow breed. Unlike lactating dairy cows, non-dairy cattle (e.g. calves or female beef cattle) will excrete less to no (e.g. males) PFAS via the milk. It is known from the literature that the fat and protein content as well as the fat composition varies

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between cow breeds (Maurice-Van Eijndhoven *et al.*, 2011). Since PFAS have an affinity for proteins and lipids, it can be assumed that the content of these substances affects the affinity of PFAS for milk and thus the amount of PFAS excreted via the milk. Therefore, the breed of a cow could also affect the elimination of PFAS via the milk;

- PFAS levels in the blood measured in either (whole) blood, blood plasma or blood serum. PFAS bind strongly to proteins and lipids in blood, which thereby help determine their accumulation and distribution in organisms (Qin *et al.*, 2024). Comparison of whole blood and plasma shows that the accumulation of compounds such as PFOA and PFOS occurs mainly in the non-cellular blood fraction. Therefore, PFAS are mainly measured in plasma or serum (Román *et al.*, 2023). Focusing solely on the 4 regulated PFAS, results expressed by volume of serum or plasma are quite similar (Ehresman *et al.*, 2007; Poothong *et al.*, 2017). For PFOS, PFOA, PFHxS and PFNA, the concentrations in plasma and serum are therefore approximately equivalent. However, for practical reasons, plasma is more suitable for collection because it can be extracted after a longer period of time if the collected blood is kept cool and not frozen, compared to the serum that needs to be extracted from clotted blood within a few hours of collection. In addition, bovine blood does not clot very well and tends to stick to the wall of blood collection tubes. It is therefore advisable to choose plasma;
- the risk of haemolysis during or after blood collection, and its effect on the PFAS concentration in plasma or serum. It is known from practice that haemolysis can occur during blood collection in cattle. This is mainly the case when blood is drawn at the tail of the animal when a lot of force has to be applied to the syringe plunger. In addition, haemolysis may occur after freezing or thawing of collected blood. During haemolysis, the cell content (low in PFAS) of the the blood is released. This leads to a dilution effect, whereby the relative concentration of PFAS in the subsequently extracted plasma or serum will be lower than when no haemolysis occurred. To avoid haemolysis as much as possible, disposable blood collection systems should be used and blood should be drawn from the carotid vein;
- error margins associated with extrapolation outside the data range of PFAS concentrations;
- uncertainty related to the limited amount of data available;
- a fairly large spread can be observed in the ratios between PFAS levels in meat and liver versus plasma, as reported in the literature. Differences in gender, duration of exposure, age and body composition may contribute to the spread in the observed ratios. In addition, no correction was made for the muscle type sampled to derive the ratios. Differences in muscle types sampled in different studies, from which the data used for the calculations were obtained, also contribute to the uncertainty and spread in the ratios found (FOVP, 2024). These uncertainty factors also apply to the large spreads in the mutual ratios between PFAS levels in the different tissues (meat, liver and kidneys). The FASFC states that they sample the diaphragm for the determination of PFAS in muscle tissue. However, this is a highly vascularised and therefore blood-rich muscle, which may cause problems in correctly estimating the ratio of PFAS levels between blood plasma on the one hand and muscle tissue (meat) on the other;
- the different half-lives published in the literature. Half-lives were reported in the literature for cattle exposed in different ways, with different washout periods and different exposure periods (FOVP, 2024). These half-lives will therefore, like the reported ratios, possibly differ between animals and it is unclear how representative they are of the half-lives of PFOS in bovine animals of Belgian operators;
- the available literature data and FASFC monitoring data on PFAS in bovines used in this opinion relate mainly to data from dairy cows and to a lesser extent to data from beef cattle. As a result, it is uncertain to what extent values derived from paired data (PFAS levels in tissues and blood of the same animal) of dairy cows are also representative of beef cattle (for which such paired data are not available in sufficient quantity).

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8. Conclusions

The SciCom assessed the possibility of establishing a guide value for PFAS compounds in bovine blood, so that it can be estimated whether the PFAS levels in meat, liver and kidneys meet the European maximum levels (Regulation (EU) 2023/915). To this end, the SciCom relied on (i) Danish Decree No. 1386 of 29/11/2023, (ii) literature data and (iii) control data of the FASFC. In addition, a linear regression analysis was applied to data of PFOS concentrations in the blood plasma and muscle tissue of the same animals (paired results for 28 cows) (Johnston *et al.*, 2023). This analysis, performed on a logarithmic transformation of the raw data, was applied to determine that a PFOS content of 0.3 µg/kg in meat (= legal European maximum level) could correspond to a most likely PFOS content of 6.2 µg/L in blood plasma with an estimated 95% confidence interval of 2.5 to 23.4 µg/L. For liver and kidneys, it was not possible to apply this type of approach due to the lack of available paired data (PFOS levels in the blood and tissues of the same animal).

The resulting lower limit of 2.5 µg/L is close to, but more conservative than, the guide value of 3.3 µg PFOS/L blood plasma applied by the Danish authorities. The Scientific Committee considers that a PFOS level of 2.5 µg/L in blood plasma could be used as a guide value for estimating conformity of PFOS levels in meat, liver and kidneys. For PFOA, PFHxS and PFNA, there are fewer or no data to derive such a possible target value in blood plasma. All in all, the Scientific Committee considers that the same approximate plasma concentration of 2.5 µg/L could also be used to estimate conformity of PFOA, PFHxS or PFNA levels in bovine meat, liver and kidneys. It should be emphasised that this guide value of 2.5 µg/L is subject to significant uncertainties (see section 7: Uncertainties) and that it is therefore not excluded that this guide value is not 100% conclusive.

Relationships between PFAS levels in bovine muscle tissue, liver and kidneys were assessed using (i) control data of the FASFC and (ii) data from the scientific literature. The purpose of this was to be able to assess whether the liver or kidneys of a slaughtered bovine should be destroyed if the meat is non-conform or vice versa. Based on the results available for the liver (n=16), it could be estimated that the PFOS concentration in the liver is on average 21 times (between 6 and 34 times) higher than that in the muscle tissue, while the PFOS concentration in the kidneys (n=8) would be on average 8 times (between 3 and 12 times) higher than that in the muscle tissue. For PFOA, PFNA and PFHxS, such a possible ratio could not be derived due to insufficient data. It should be noted that these ratios for PFOS were determined on the basis of a relatively limited number of data from different studies with different experimental designs, and that no distinction was made between breed, sex and lactation status of the cattle when deriving the ratios (see section 7. Uncertainties).

Finally, SciCom has evaluated the possibility of determining which, after removal of the source of contamination, would be the most favourable time to perform an ante-mortem blood test to estimate that bovine muscle tissue, liver and kidneys will be compliant for PFAS. To this end, literature sources on the depletion kinetics of PFAS in bovine blood were studied. For PFOS, elimination half-lives in blood ranged from 39 (lactating cow) to 120 days (steer). For PFOA, elimination half-lives in blood were 1.3 (lactating cow) and 19.2 days (steers). For PFNA, the elimination half-lives in blood were 8.7 (lactating cow) and 12.3 days (beef cattle), and for PFHxS, the half-life was 9.3 days for beef cattle.

Chou *et al.* (2023) have implemented a physiologically based pharmacokinetic (PBPK) model in an interactive generic platform, publicly accessible via the internet, where tissue withdrawal times for PFOA, PFOS and PFHxS can be calculated for beef cattle and dairy cows. An important limitation of the platform is that different parameter values must be known to be filled in in the application and that it is not possible to introduce the removal of the source of contamination in the model. In addition, other equations that would approximate the depletion kinetics in blood plasma or blood serum are reported in scientific publications (Johnston *et al.*, 2023; Mikkonen *et al.*, 2023), but these also appear to be

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little useful in practice due to the required parameters that must be known and the complexity. Therefore, the working group itself developed a pragmatic approach that could be useful for calculating a withdrawal time after which a second blood sample should be taken and analysed, in the event of an initial blood test indicating a too high level of PFAS. The equation for calculating such a withdrawal time (see section 9. Recommendations below) contains as constants the guide value for PFAS in blood plasma of 2.5 µg/L and an elimination half-life for PFAS (for which a value is to be selected from the literature). Therefore, a withdrawal time calculated in this way is subject to significant uncertainties.

9. Recommendations

When a PFAS contamination is identified on a farm, it is recommended to identify and eliminate the source of contamination (e.g. water, soil or feed) or to prevent animals from having access to it . The Scientific Committee formulates the threshold value of 2.5 µg/L bovine blood plasma as a possible guide value for ante-mortem assessment of conformity with the legal European maximum levels of PFOS, PFOA, PFHxS or PFNA in meat, liver and kidneys. As discussed above, this guide value of 2.5 µg/L is subject to significant uncertainties. The Scientific Committee stresses that it cannot be excluded that potential non-compliant results of PFAS in meat, liver or kidneys of bovines can be observed after a blood test meeting the guide value of 2.5 µg/L.

The Scientific Committee therefore recommends evaluating in the near future in how many cases a level of PFOS, PFOA, PFHxS or PFNA in blood plasma that does not exceed the guide value of 2.5 µg/L is actually related to the conformity of meat, kidneys and liver for these 4 PFAS. It is therefore recommended to determine and collect paired data (from the same animal) on PFAS in the blood plasma, muscle tissue, liver and kidneys of bovines from farms affected by a PFAS contamination. This additional data can then serve as a basis for confirming or adjusting the guide value of 2.5 µg/L if necessary.

If an analysis of bovine blood plasma reveals that the concentration of PFOS, PFOA, PFHxS or PFNA is higher than the guide value of 2.5 µg/L, it is recommended to perform a new blood plasma analysis at a later time until a result lower than this value is obtained. This later time can be calculated on the basis of the following equation:

$$withdrawal\ time = \frac{(\log C_0 - \log 2{,}5) \times 2{,}303}{(0{,}693/DT_{50})}$$

Equation of the withdrawal time (t, in days) in blood plasma/serum to obtain the targeted guide value, where C_0 represents the starting concentration (µg/L) of PFOS, PFOA, PFHxS or PFNA in the blood plasma/serum, i.e. the concentration measured at the first sampling; 2.5 µg/L the proposed blood plasma/serum concentration guide value; DT_{50} the plasma/serum elimination half-life of PFOS, PFOA, PFHxS or PFNA (days).

The Scientific Committee also refers to the uncertainties discussed above. The withdrawal time should therefore be seen as an indication for determining a time when a second blood test may be useful. However, it cannot be ruled out that after the estimated withdrawal time, a bovine would still show a PFAS level in the blood plasma higher than 2.5 µg/L. In addition, calculated withdrawal times are expected to be rather long based on the relatively long elimination half-lives of PFAS described in literature. For these reasons, the practical benefit of these withdrawal times is rather low.

In addition, it is recommended for PFOS to use the following ratios in order to be able to estimate its relative concentration in bovine muscle tissue, liver or kidneys without having to measure all these

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tissues separately: the PFOS concentration in the liver would be on average 21 times (between 6 and 34 times) higher than that in the muscle tissue, while the PFOS concentration in the kidneys would be on average 8 times (between 3 and 12 times) higher than that in the muscle tissue. The uncertainties that apply to these ratios are discussed above. The Scientific Committee recommends that, in the near future, paired samples (from the same animal) of muscle tissue, liver and kidneys from Belgian bovine animals be taken to verify these PFOS ratios between the different animal tissues.

On behalf of the Scientific Committee,
The Chair,

Dr L. Herman
Brussels, 17/01/2025

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References

Chou, W.C., Tell, L.A., Baynes, R.E., Davis, J.L., Cheng, Y.-H., Maunsell, F. P., Riviere, J. E., & Lin, Z. (2023). Development and application of an interactive generic physiologically based pharmacokinetic (igPBPK) model for adult beef cattle and lactating dairy cows to estimate tissue distribution and edible tissue and milk withdrawal intervals for per- and polyfluoroalkyl substances (PFAS). *Food and Chemical Toxicology* 181, 114062. https://doi.org/10.1016/j.fct.2023.114062

Death, C, Bell, C., Champness, D. Milne, C., Reichman, S., & Hagen, T. (2021). Per- and polyfluoroalkyl substances (PFAS) in livestock and game species: A review. *Science of the Total Environment* 774, 144795. https://doi.org/10.1016/j.scitotenv.2020.144795 (+ supplementary data: https://ars.els-cdn.com/content/image/1-s2.0-S0048969720383285-mmc1.docx)

Danish Decree No 1386 of 29/11/2023: Guide values for a maximum PFOS concentration in bovine and ovine serum. https://www.retsinformation.dk/eli/lta/2023/1386

Drew, R., Hagen, T., G., Champness, D., & Sellier, A. (2021): Half-lives of several polyfluoroalkyl substances (PFAS) in bovine serum and tissues. *Food additives & contaminants: A* unit. https://doi.org/10.1080/19440049.2021.1991004

Ehresman, D. J., Froehlich, J. W., Olsen, G. W., Chang, South, & Butenhoff, J. L. (2007). Comparison of human whole blood, plasma, and serum matrices for the determination of perfluorooctanesulfonate (PFOS), perfluorooctanoate (PFOA), and other fluorochemicals. *Environmental Research* 103, 176-184. https://doi.org/10.1016/j.envres.2006.06,008

Front Office Food and Product Safety (FOVP) (2024). Assessment of PFOS kinetics in lamb, sheep and bovine animals. https://www.rivm.nl/documenten/beoordeling-over-overdracht-en-kinetiek-van-pfos-in-lam-schaap-en-rund

Hoogenboom *et al.* (2024). Manuscript in preparation.

Johnston, J.J., Ebel, E.D., Williams, M. S., Esteban, E., Lupton, S. J., Scholljegerdes, E. J., Ivey, S. L., Powell, M. R., & Smith, D.J. (2023). Blood-based ante-mortem method for estimating PFOS in beef from contaminated dairy cattle. *ACS Agricultural Science & Technology* 3, 835-844. https://doi.org/10.1021/acsagscitech.3c00102 (+ supporting information: https://pubs.acs.org/doi/suppl/10.1021/acsagscitech.3c00102/suppl_file/as3c00102_si_001.pdf

Kowalczyk, J., Ehlers, S., Forest, P., Schafft, H., & Lahrssen-Wiederholt, M. (2012). Transfer of perfluorooctanic acid (PFOA) and perfluorooctane sulfonate (PFOS) from contaminated feed into milk and sheepmeat: Pilot study. *Arch Environ Contam Toxicol* 63, 288-298. https://doi.org/10.1007/s00244-012-9759-2

Kowalczyk, J., Ehlers, S., Oberhausen, A., Tablesman, M., Forest, P., Schafft, H., & Lahrssen-Wiederholt, M. (2013). Absorption, distribution, and milk secretion of the perfluoroalkyl acids PFBS, PFHxS, PFOS, and PFOA by dairy cows fed naturally contaminated feed. Journal of Agricultural and Food Chemistry. 61(12):2903–2912. https://doi.org/10.1021/jf304680j

Lupton, S. J., Huwe, J. K., Smith, D. J. Dearfield, K. L., & Johnston, J. J. (2012). Absorption and excretion of ^{14} C-perfluoro-actanoic acid (PFOA) in Angus cattle (*Bos taurus*). *Journal of Agricultural and Food Chemistry* 60(4), 1128-1134. https://doi.org/10.1021/jf2042505

29401399, 2025, 2, Downloaded from https://efsa.onlinelibrary.wiley.com/doi/10.2903/fr.efsa.2025.FR-0056 by Cochrane France, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Lupton, S. J., Huwe, J. K., Smith, D.J., Dearfield, K.L., & Johnston, J.J. (2014). Distribution and excretion of perfluorooctane sulfonate (PFOS) in beef cattle (*Bos taurus*). *Journal of Agricultural and Food Chemistry* 62(5), 1167-1173. https://doi.org/10.1021/jf404355b

Lupton, S. J., Dearfield, K. L., Johnston, J. J., Wagner, S., & Huwe, J.K. (2015). Perfluorooctane Sulfonate Plasma Half-Life Determination and Long-Term Tissue Distribution in Beef Cattle (*Bos taurus*). *Journal of Agricultural and Food Chemistry* 63(51), 10988-10994. https://doi.org/10.1021/acs.jafc.5b04565

Maurice-Van Eijndhoven, M. H. T., Hiemstra, S. J., & Calus, M. P. L. (2011). Short communication: Milk fat composition of 4 cattle breeds in the Netherlands. *J. Dairy Sci.* 94, 1021-1025. https://doi.org/10.3168/jds.2009-3018

Mikkonen, A. T., Martin, J., Upton, R. N., Moenning, J.-L., Numata, J., Taylor, M. P., Roberts, M. S., & Mackenzie, L. (2023). Dynamic exposure and body load models for per- and polyfluoroalkyl substances (PFAS) enable management of food safety risks in cattle. *Environment International* 180, 108218. https://doi.org/10.1016/j.envint.2023.108218

Mogensen, U. B., Grandjean, P., Nielsen, F., Weihe, P., & Budtz-Jørgensen, E. (2015). Breastfeeding as an exposure pathway for perfluorinated alkylates. *About. Saw. Tech* 49 (17), 10466-10473. https://doi.org/10.1021/acs.est.5b02237

Poothong, S., Thomsen, C., Padilla-Sanchez, J.A., Papadopoulou, E., & Haug, L. S. (2017). Distribution of novel and well-known poly- and perfluoroalkyl substances (PFASs) in human serum, plasma, and whole blood. *Enivron. Saw. Technol.* 51, 13388-13396. https://doi.org/10.1021/acs.est.7b03299

Qin, W., Escher, B. I., Huchthausen, J., Fu, Q., & Henneberger, L. (2024). Species difference? Bovine, trout, and human plasma protein binding of per- and polyfluoroalkyl substances. *Environmental Science & Technology* 58, 9954-9966. https://doi.org/10.1021/acs.est.3c10824

Román, A. S., Abilleira, E., Irizar, A., Santa Marina, L., Gonzalez-Gaya, B., & Etxebarria, N. (2023). Optimization for the analysis of 42 per- and polyfluorinated substances in human plasma: a high-throughput method for epidemiological studies. *Journal of Chromatography A* 1712, 464481. https://doi.org/10.1016/j.chroma.2023.464481

van Asselt, E. D., Kowalczyk, J., Van Eijkeren, J. C. H., Zeilmaker, M. J., Ehlers, S., Forest, P., Lahrssen-Wiederholt, M., & Van Der Fels-Klerx, H. J. (2013). Transfer of perfluorooctane sulfonic acid (PFOS) from contaminated feed to milk. *Food chemistry* 141(2), 1489-1495. https://doi.org/10.1016/j.foodchem.2013.04,035

Vestergren, R., Orata, F., Berger, U., & Cousins, I. T. (2013). Bioaccumulation of perfluoroalkyl acids in dairy cows in a naturally contaminated environment. *Environmental Science and Pollution Research* 20(11), 7959-7969. https://doi.org/10.1007/s11356-013-1722-x

Vinggaard, A. M. (2021). *Transfer of PFOS from water and grass to cattle and sheep.* DTU National Food Institute. https://backend.orbit.dtu.dk/ws/files/268384181/Overf_rsel_af_PFOS_fra_vand_og_gr_s_til_kv_g_og_f_r.pdf

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Presentation of the Scientific Committee established at the FASFC

The Scientific Committee is an advisory body established at the Belgian Federal Agency for the Safety of the Food Chain (FASFC) that provides **independent scientific opinions** on risk assessment and risk management in the food chain, at the request of the Chief Executive Officer of the FASFC, the Minister competent for food safety or at its own initiative. The Scientific Committee is administratively and scientifically supported by the Staff direction for Risk Assessment of the Agency.

The Scientific Committee consists of 22 members who are appointed by Royal Decree on the basis of their scientific expertise in areas related to the safety of the food chain. When preparing an opinion, the Scientific Committee may call upon external experts who are not a member of the Scientific Committee. Similar to the members of the Scientific Committee, they must be able to work independently and impartially. To ensure the independence of the opinions, potential conflicts of interest are managed transparently.

The opinions are based on a scientific assessment of the question. They express the view of the Scientific Committee, which is taken in consensus on the basis of a risk assessment and the existing knowledge on the subject.

The opinions of the Scientific Committee may contain **recommendations** for food chain control policy or for the stakeholders. The follow up of these recommendations for policy is the responsibility of the risk managers.

Questions related to an opinion can be directed to the secretariat of the Scientific Committee: Secretariaat.SciCom@favv.be.

Members of the Scientific Committee

The Scientific Committee is composed of the following members:

A. Clinquart[9], P. Delahaut, B. De Meulenaer, N. De Regge, J. Dewulf, L. De Zutter, A. Geeraerd Ameryckx, N. Gillard, L. Herman, K. Houf, N. Korsak, L. Maes, M. Mori, A. Rajkovic, N. Roosens, C. Saegerman, M.-L. Scippo, P. Spanoghe, K. Van Hoorde, Y. Vandenplas[10], F. Verheggen, P. Veys[11], S. Vlaeminck

Conflicts of interest

No conflicts of interest were identified.

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The Scientific Committee would also like to thank J. Dewulf and N. Roosens for the 'deep reading' of the opinion.

[9] member until December 2021
[10] member until September 2023
[11] member from January 2022

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Composition of the working group

The working group was composed of:

Members of the Scientific Committee:	M.-L. Scippo (reporter), C. Saegerman, B. De Meulenaer, N. Gillard
External experts:	E. Van Pamel (ILVO), L. Joly (Sciensano), L. Bervoets (UAntwerp), S. Croubels (UGent), G. Eppe (ULiège), L. Pussemier
File Manager:	K. Houben, O. Rojo Poveda

The following members of the administration attended the activities of the working group (as observers): R. Regnier and E. Moons of the Federal Agency for the Safety of the Food Chain.

Legal framework

Act of 4 February 2000, on the creation of the Federal Agency for the Safety of the Food Chain, in particular Article 8;

The Royal Decree of 19 May 2000, on the composition and operating procedures of the Scientific Committee, as established at the Federal Agency for the Safety of the Food Chain;

The Internal Rules as mentioned in Article 3 of the Royal Decree of 19 May 2000, on the composition and operating procedures of the Scientific Committee, as established at the Federal Agency for the Safety of the Food Chain, approved by the Minister on 24 September 2020.

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The Scientific Committee at all times reserves the right to modify the opinion should new information and data become available after the publication of this version of the opinion.

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Appendix 1: Machine translation into English of "Vinggaard, A. M. (2021). *Overførsel af PFOS fra vand og græs til kvæg og får.* DTU National Food Institute" (translation by www.onlinedoctranslator.com)

Transfer of PFOS from water and grass to cattle and sheep

Vinggaard, Anne Marie

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Transfer of PFOS from water and grass to cattle and sheep

Background

FVST is increasingly receiving inquiries from municipalities, organizations and citizens regarding contamination of grass and water with PFOS in areas near former firefighting training sites. There is a demand for ways to estimate PFOS levels in cattle and sheep grazing in the area and/or drinking from the water.

This task was completed with a relatively short response time. In addition, our knowledge about the uptake and accumulation of PFOS in cattle and sheep is limited and principles are transferred from human toxicokinetics.

Conclusion

Recent studies have been conducted in cattle and sheep concerning the uptake of PFOS from water and silage, which show how the substance is distributed and excreted in the animals. Firstly, so-called transfer factors are given, which say something about how much PFOS is transferred from water/silage to serum (Table 1). Secondly, distribution coefficients are given for the ratio between serum/plasma and, for example, muscle tissue in cattle and sheep, which is primarily what we eat (Table 2). Finally, distribution coefficients between serum and milk are given, so that the PFOS level in cow's milk can be estimated (Table 3). Finally, we have made a calculation example for how to convert from PFOS levels in drinking water to PFOS levels in meat and milk.

PFOS uptake in cattle and sheep from drinking water and silage

A thorough study from Australia (Drew et al. 2021) has recently been published, in which the uptake of PFOS in cattle and sheep from contaminated drinking water has been investigated. The study has been run over a long period of time, which means that steady state has been reached, and this is important for the reliability of the data. The results for PFOS show that the so-called 'transfer factor', i.e. the ratio between the content in serum relative to the content in drinking water, is 140 in adult cattle that are not pregnant and are not dairy cattle. For ewes, the corresponding transfer factor is 20.

These transfer factors apply if the animals have been exposed to the contaminated water or feed for at least 3-5 half-lives. This corresponds to 222-370 days in cattle (T½=74 days) and 195 – 325 days in sheep (T½=65 days). This is the case in this study. If the animals have been exposed for a shorter time, the transfer factors will be overestimated.

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Table 1: Transfer factors from drinking water/silage PFOS levels to serum levels in cattle and sheep

Source	Animal	Receive compartment-intended	Transfer factor (serum conc/water conc)	Reference-ence	Comment
Drink-kevand	Adult cattle (non-pregnant, non-lactating gutter)	Serum	**140**	Drew it all 2021	Steady state has been reached
Drink-kevand	Ewe (non-drag-tig, non-lacquer-tering)	Serum	**20**	Drew it all 2021	Steady state has been reached
Ensi-build/build	Adult cattle (non-pregnant, non-lactating gutter)	Serum (converted from whole blood: serum level is twice as high as whole blood level)	**30**	Western branch one all 2013	Levels in the feed are much lower than in drinking water in Drew et al 2021 – may have affected TF

Transfer of PFOS from plasma to tissues in cattle and sheep

Partition coefficients for PFOS from tissue to plasma are given in Table 2. Since PFAS levels in serum and plasma can be set to 1:1, these factors can be used to calculate PFOS levels in muscle tissue, liver and kidney for a specific contamination of serum.

Measurements of the liver can be a good supplement to measurements in muscle tissue, partly because consumers of this organ will be particularly exposed, and partly because measurements of the liver will lower the limit for when contaminated animals can be identified, even if the content in the muscle tissue may be below the LOQ of the method. In dairy cattle, the PFOS level in muscle (i.e. meat) is 1/100 of the concentration in the liver (Death et al 2021).

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Table 2: Tissue (µg/kg) to plasma (µg/L) partition coefficients for PFOS in cattle and sheep(modified from Death et al 2021)

Animal	PFOS			Reference
	Muscle	Liver	Kidney	
Sheep	**0.15**	5.1	1.1	Kowalczyk et al 2012*
Cattle	**0.07**			LANUV 2011#
	0.07	1.6	0.6	Kowalczyk et al 2013
	0.09	0.6b		Vestergren et al 2013
	0.02	0.3	0.07	Lupton et al 2014**
	0.03	1.0	0.6d	Lupton et al 2015**

See main article for references

*Data from two ewes; steady-state assumed not to have been achieved

#Data from three fattened bulls.

**Steady-state assumed not to have been achieved

bMean PFOS concentrations in whole blood (converted from ng/kg to ng/L using a specific gravity of 1.06 from Trudnowski and Rico (1974), liver, muscle, and milk were 117 ± 20 (234 ng/L in serum), 130 ± 32, 21 ± 19 ng/kg, and 6.2 ± 1.1 ng/L, respectively.

dIn steers on day 343, average PFOS concentrations in plasma, muscle (avg. of shoulder, tenderloin, ribeye, and rump), liver, kidney, and fat (avg. of back and IP fat) were 0.15 µg/mL, 0.005, 0.149, 0.082, and 0.026 µg/g, respectively. In heifers on day 105, average (n=2) PFOS concentrations were 39.24 µg/mL, 0.835, 8.76, 3.96, and 1.095 µg/g, respectively. In heifers on day 343, average (n=2) PFOS concentrations were 8.33 µg/mL, 0.283, 4.74, 2.36, and 0.6 µg/g, respectively. Tissue to serum partition coefficients therefore:

- For heifers, 0.02 and 0.03 for muscle; 0.2 and 0.6 for liver; 0.1 and 0.3 for kidney. For
- steers, 0.03 for muscle, 1 for liver, and 0.6 for kidney.

Transfer of PFOS from serum to milk in cattle and sheep

Table 3: Transfer factors for serum to milk in cattle and sheep

Source	Animal	Receiving compartment	Transfer factor (milk conc/serum conc)	Reference
Serum	Milking wall	Milk	**0.013**	Kowalczyk 2012, 2013
Serum	Sheep	Milk	**0.06**	Kowalczyk 2012, 2013

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A calculation example

A water sample containing**10 ng/L PFOS**and it is desired to estimate how much PFOS can be expected in meat and milk from cattle that have drunk the water for at least 1 year.

PFOS serum concentration in cattle:	10 ng/L x 140	= 1400 ng/L
Min PFOS concentration in beef (muscle):	1400 ng/L x 0.02	= 28 ng/L
Max PFOS concentration in beef (muscle):	1400 ng/L x 0.09	= 126 ng/L
Min PFOS concentration in beef (liver): Max	1400 ng/L x 0.3	= 420 ng/L
PFOS concentration in beef (liver): PFOS	1400 ng/L x 1.6	= 2240 ng/L
concentration in cow's milk:	1400 ng/L x 0.013	= 18 ng/L

The corresponding calculations for sheep will be:		
PFOS serum concentration in sheep:	10 ng/L x 20	= 200 ng/L
PFOS concentration in sheep meat (muscle):	200 ng/L x 0.15	= 30 ng/L
PFOS concentration in sheep milk:	200 ng/L x 0.06	= 12 ng/L

If the PFOS content in a grass sample is to be converted, it is suggested to use a transfer factor of 30 for cattle. This value has been measured for uptake via silage and barley and it is assumed that uptake from grass is not significantly different from that for this feed.

Accumulation of PFOS from drinking water in sheep is significantly lower than in cattle. Since the half-lives are not that different in cattle and sheep, this is probably because sheep drink less water per kg of body weight than cattle do.

The above calculations should only be seen as a rough guideline. In addition to the general methodological uncertainties that will always be present in the studies that have calculated the value of the transfer factors, it should be remembered that the content in the meat reflects the animals' exposure over several months. The content in the animals' feed and drinking water will naturally vary during the time that the animals are in the contaminated areas, and this variation is not seen unless several measurements have been made from start to finish during the period that the animals have grazed on the area. In addition, animals may have different preferences for where in the field they move (and eat) and the contamination in the area will usually hardly be evenly distributed.

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References

Death C, Bell C, Champness D, Milne C, Reichman S, Hagen T. Per- and polyfluoroalkyl substances (PFAS) in livestock and game species: A review. Science of The Total Environment, Volume 774, 2021.

Drew R, Hagena TG, Champness D. Accumulation of PFAS by livestock – determination of transfer factors from water to serum for cattle and sheep in Australia. Food Additives & Contaminants: Part A, 2021. DOI: 10.1080/19440049.2021.1942562

Kowalczyk J, Ehlers S, Fürst P, Schafft H, Lahrssen-Wiederholt M. Transfer of Perfluorooctanoic Acid (PFOA) and Perfluorooctane Sulfonate (PFOS) From Contaminated Feed Into Milk and Meat of Sheep: Pilot Study. Arch Environ Contam Toxicol (2012) 63:288–298.

Kowalczyk J, Ehlers S, Oberhausen A, M Tischer, Fürst P, Schafft H, Lahrssen-Wiederholt M. Absorption, Distribution, and Milk Secretion of the Perfluoroalkyl Acids PFBS, PFHxS, PFOS, and PFOA by Dairy Cows Fed Naturally Contaminated Feed. J. Agric. Food Chem. 2013, 61, 12, 2903.

Vestergren R, Orata F, Berger U & Cousins IA. Bioaccumulation of perfluoroalkyl acids in dairy cows in a naturally contaminated environment. Environ Sci Pollut Res (2013) 20:7959–7969.

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Appendix 2: DTU National Food Institute (2021). *Overførsel af PFAS fra vand og græs til kvæg*

Overførsel af PFAS fra vand og græs til kvæg

Baggrund

FVST får i stigende grad henvendelser fra kommuner, organisationer og borgere vedrørende forurening af græs og vand med PFAS. Der bliver efterspurgt måder at estimere PFAS niveauet i kvæg der græsser på området og/eller drikker af vandet.

Det drejer sig om en pakke på 12 fluorstoffer, som typisk indgår i analysen, hvor FVST indtil nu har set fund af dem alle med undtagelse af de to sidstnævnte.
PFCA (C4-C10): PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA,
PFSA (C4-C9): PFBS, PFHxS, PFOS,
Andre: PFOSA, 6:2 FTS.

FVST vil forhøre om:

1. DTU kan foreslå os tilsvarende værdier for acceptable niveauer i græs og vand (for optag i kvæg, får og fisk), når det gælder for **summen af fluorstoffer** og ikke kun PFOS alene?
2. DTU kan vejlede os i, hvordan vi skal vurdere fund i fødevarer af andre fluorstoffer end de fire stoffer, hvor EU har foreslået grænseværdier?

DTU må korrigere en formulering i opgaven. Det må pointeres at DTU ikke har leveret acceptable indhold i fødevarer for PFOS til FVST som angivet i spørgsmålet. FVST har på basis af data leveret fra DTU og LOQ ("limit of quantification") for PFOS fastsat acceptable indhold af PFOS i vand og foder. Dette er risikohåndtering og uden for DTU's resortområde.
Denne opgave er udført med relativ kort svarfrist og vi har derfor foretaget en foreløbig konservativ vurdering af de fluorstoffer, hvor datagrundlaget er mangelfuldt eller ikke eksisterende og sidestillet dem med PFOS. Derudover er vores viden omkring optag og akkumulering af PFAS i kvæg begrænset og principper er overført fra human toksikokinetik. I dette svar er der alene fokus på kvæg.

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Konklusion

Generelt er de stillede spørgsmål meget omfattende og vil kræve flere måneders arbejde, og det er usikkert om der overhovedet er de nødvendige data til at levere et fyldestgørende svar. Specielt er det sidste spørgsmål vedrørende risikovurderinger af de øvrige PFAS en længerevarende opgave i sig selv.

Ad 1. Det korte svar er nej. Vi kan ikke angive forslag til acceptable grænseværdier i græs og vand for summen af 12 fluorstoffer. EFSA har udelukkende risikovurderet og angivet TWI for 4 fluorstoffer (4,4 ng/kg kropsvægt per uge for PFOA, PFNA, PFOS og PFHxS) og de resterende har ikke tilsvarende grænseværdier for et tolerabelt indtag. Derudover kan der ikke findes transfer-faktorer for alle 12 PFAS.
Dog findes der transfer-faktorer fra drikkevand til serum for 4 andre PFAS udover PFOS. Dog mangler vi generelt data for de kortkædede PFAS. Men der findes ikke grænseværdier for de 8 PFAS og derfor kan der ikke fastsættes grænseværdier på basis af en sundhedsfaglig vurdering.

Ad 2. De øvrige PFAS (ud over de såkaldte 'EFSA4': PFOA, PFNA, PFOS og PFHxS) er os bekendt ikke risikovurderede endnu og grænseværdier er ikke fastsat. Det ligger uden for denne opgaves rammer at risikovurdere disse 8 PFAS. Der foreligger sammenlignelige 28-dages forsøg på rotter for adskillige PFCA og PFSA udført af NTP i USA og på basis af disse rapporter har RIVM fastsat 'Relative Potens Faktorer' for en række PFAS (publikation vedhæftet). Resultaterne herfra kan eventuelt anvendes til at sammenligne toksicitet for de enkelte PFAS med forskellige kædelængder. Dog er der meget der tyder på at brug af data fra gnavere i risikovurderingen vil underestimere risikoen til PFAS i mennesker. En vurdering af dette er et omfattende arbejde, der ligger uden for trækningsrettens rammer.

Beregning af PFAS optag i kød

PFAS indholdet kan estimeres ved hjælp af en metode, hvor transfer-faktorer anvendes i første trin til at estimere PFAS optag i serum fra fx drikkevand. Vi fandt transfer-faktorer for 5 PFAS fra drikkevand til serum i kvæg (Tabel 1). I næste trin anvendes fordelingskoefficienter mellem serum og væv (fx muskel som er det vi typisk spiser) til at vurdere den endelige koncentration i kødet. Vi fandt fordelingskoefficienter for 3 PFAS (Tabel 2).

Studiet der rapporterer om transfer-faktorer fra kontamineret drikkevand til kvæg er meget grundigt (Drew et al. 2021). Studiet er kørt over lang tid, hvilket betyder at steady state er nået, og dette er vigtigt for datas pålidelighed. Resultaterne for PFAS viser den såkaldte 'transfer-faktor' dvs ratio mellem indholdet i serum relativt til indholdet i drikkevandet i voksent kvæg, som ikke er drægtigt og ikke er malkekvæg. Disse transfer-faktorer gælder hvis dyrene har været eksponerede for det kontaminerede vand eller foder i mindst 3-5 halveringstider. Dette svarer til 222- 370 dage i kvæg (T½=74 dage) og er

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tilfældet i dette studie. Hvis dyrene har været eksponerede i kortere tid, vil transfer faktorerne være overestimerede.

Tabel 1. Transfer-faktorer for PFAS fra drikkevand (Drew et a. 2021) **eller ensilage/byg** (Vestergren et al. 2013) **til serum i kvæg**. *Transfer-faktorer for kvæg blev beregnet ved at dividere steady state serum-koncentrationen (ng PFAS/mL) med vandkoncentrationen (ng PFAS/mL).*

Kilde	Dyr	Modtage kompartment	Transfer-faktorer (gennemsnits-niveau) (serum konc/vand konc)					
			PFHxS	PFHpS	PFOS	PFOA	PFNA	PFDA
Drik-kevand	Voksen kvæg (ikke-drægtig, ikke-lakte-rende)	Serum	65	130	140		170	120
Ensilage/ byg	Voksen kvæg (ikke-drægtig, ikke-lakte-rende)	Serum	-	-	30 [a]	~0.4[a]	~12[a]	~10[a]

a: tallene er aflæst fra figur 2 i Vestergren el al. 2013.

I tabel 2 er fordelingskoefficienter for PFOS, PFOA og PFHxS fra væv til plasma angivet. Da PFAS niveauer i serum og i plasma kan sættes til 1:1, kan disse faktorer bruges til at beregne PFAS niveauet i muskelvæv og lever ved en specifik kontaminering af serum. Data fra Kowalczyk et al. 2013 hvor både PFOS, PFOA og PFHxS er undersøgt indikerer at fordelingskoefficienterne er relative sammenlignelige for de tre stoffer (0.07, 0.07 og 0.05).

Tabel 2. Fordelingskoeeficienter for væv til serum for PFAS i kvæg (modificeret fra Death et al 2021)

Species	Study	**PFOS**		**PFOA**		**PFHxS**	
		Muskel	Lever	Muskel	Lever	Muskel	Lever
Cattle	*LANUV 2011*[#]	0.07					
	Kowalczyk et al 2013	**0.07**	1.6	**0.07**	1.2	**0.05**	0.16
	Vestergren et al 2013	0.09	0.6[b]	0.2	0.3[c]		
	*Lupton et al 2014***	0.02	0.3				
	*Lupton et al 2015***	0.03	1.0				

[#] Data from three fattened bulls; ** steady-state assumed not to have been achieved; [b] Mean PFOS concentrations in whole blood (converted from ng/kg to ng/L using a specific gravity of 1.06 from Trudnowski and Rico (1974), liver, muscle, and milk were 117 ± 20 (234 ng/L in serum), 130 ± 32, 21 ± 19 ng/kg, and 6.2 ± 1.1 ng/L, respectively.

Beregningsmetoder

PFOS serum-koncentration i kvæg: 2 ng/L x 140 = 280 ng/L
Min PFOS koncentration i oksekød (muskel): 280 ng/L x 0.02 = **5.6 ng/L ~ 6.0 ng/kg kød**
Max PFOS koncentration i oksekød (muskel): 280 ng/L x 0.09 = **25.2 ng/L~ 27.1 ng/kg kød**

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Estimering af transfer-faktorer for andre PFAS

For andre PFAS uden data for transfer faktorer har vi valgt en konservativ tilgang og foreslår derfor at de tildeles samme værdier som PFOS, som er worst-case med en enkelt undtagelse (Tabel 1).

Usikkerheder

Ovenstående beregninger skal alene ses som en grov rettesnor. Ud over de almindelige metodiske usikkerheder der altid vil være i de undersøgelserne der har beregnet værdien af transfer-faktorerne, så skal man huske at indholdet i kødet afspejler dyrenes eksponering over flere måneder. Indholdet i dyrenes foder og i drikkevand vil naturligt variere i løbet af den tid dyrene befinder sig på de kontaminerede områder, og denne variation ser man ikke, medmindre man har foretaget adskillige målinger fra start til slut i den periode hvor dyrene har græsset på arealet. Hertil kommer at dyr kan havde forskellige præferencer for, hvor på marken de færdes (og spiser) og kontamineringen på området vil som regel næppe være jævnt fordelt.

Analyser af lever som supplement

Målinger i muskelvæv bør suppleres med målinger i leveren, dels fordi forbrugere af dette organ vil være særligt eksponeret, dels fordi målinger af leveren vil sænke grænsen for hvornår man kan identificere kontaminerede dyr, selvom indholdet i muskelvævet måtte ligge under metodens LOQ. I malkekvæg er PFOS niveauet i muskler (dvs kød) 1/100 af koncentrationen i leveren (Death et al 2021).

Referencer

Death C, Bell C, Champness D, Milne C, Reichman S, Hagen T. Per- and polyfluoroalkyl substances (PFAS) in livestock and game species: A review. Science of The Total Environment, Volume 774, 2021.

Drew R, Hagena TG, Champness D. Accumulation of PFAS by livestock – determination of transfer factors from water to serum for cattle and sheep in Australia. Food Additives & Contaminants: Part A, 2021. DOI: 10.1080/19440049.2021.1942562

Kowalczyk J, Ehlers S, Fürst P, Schafft H, Lahrssen-Wiederholt M. Transfer of Perfluorooctanoic Acid (PFOA) and Perfluorooctane Sulfonate (PFOS) From Contaminated Feed Into Milk and Meat of Sheep: Pilot Study. Arch Environ Contam Toxicol (2012) 63:288–298.

Kowalczyk J, Ehlers S, Oberhausen A, M Tischer, Fürst P, Schafft H, Lahrssen-Wiederholt M. Absorption, Distribution, and Milk Secretion of the Perfluoroalkyl Acids PFBS, PFHxS, PFOS, and PFOA by Dairy Cows Fed Naturally Contaminated Feed. J. Agric. Food Chem. 2013, 61, 12, 2903.

Vestergren R, Orata F, Berger U & Cousins IA. Bioaccumulation of perfluoroalkyl acids in dairy cows in a naturally contaminated environment. Environ Sci Pollut Res (2013) 20:7959–7969.

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Appendix 3: Machine translation into English of "DTU (2021). *Overførsel af PFAS fra vand og græs til kvæg"* (translation by www.onlinedoctranslator.com)

Transfer of PFAS from water and grass to cattle

Background

FVST is increasingly receiving inquiries from municipalities, organizations and citizens regarding contamination of grass and water with PFAS. There is a demand for ways to estimate PFAS levels in cattle grazing in the area and/or drinking from the water.

This concerns a package of 12 fluorine substances, which are typically included in the analysis, where FVST has so far seen findings of all of them with the exception of the last two.

PFCA (C4-C10): PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFSA
(C4-C9): PFBS, PFHxS, PFOS,
Other: PFOSA, 6:2 FTS.

FVST will inquire about:

1. DTU can suggest to us equivalent values for acceptable levels in grass and water (for uptake in cattle, sheep and fish) when it comes to**the sum of fluorine substances**and not just PFOS alone?
2. Can DTU guide us on how to assess findings in food of fluorine substances other than the four substances for which the EU has proposed limit values?

DTU must correct a wording in the assignment. It must be pointed out that DTU has not provided acceptable levels in food for PFOS to FVST as stated in the question. FVST has, on the basis of data provided by DTU and the LOQ (limit of quantification) for PFOS, determined acceptable levels of PFOS in water and feed. This is risk management and outside DTU's remit.

This task was carried out with a relatively short response time and we have therefore made a preliminary conservative assessment of the fluorine substances for which the data basis is inadequate or non-existent and equated them with PFOS. In addition, our knowledge about the uptake and accumulation of PFAS in cattle is limited and principles are transferred from human toxicokinetics. In this response, the focus is solely on cattle.

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Conclusion

In general, the questions asked are very extensive and will require several months of work, and it is uncertain whether there is even the necessary data to provide a complete answer. In particular, the last question regarding risk assessments of the other PFAS is a long-term task in itself.

Re 1. The short answer is no. We cannot provide proposals for acceptable limit values in grass and water for the sum of 12 fluorinated substances. EFSA has only assessed the risk and stated TWI for 4 fluorinated substances (4.4 ng/kg body weight per week for PFOA, PFNA, PFOS and PFHxS) and the remaining ones do not have corresponding limit values for a tolerable intake. In addition, transfer factors cannot be found for all 12 PFAS.

However, there are transfer factors from drinking water to serum for 4 other PFAS besides PFOS. However, we generally lack data for the short-chain PFAS. However, there are no limit values for the 8 PFAS and therefore limit values cannot be set on the basis of a health assessment.

Ad 2. The other PFAS (in addition to the so-called 'EFSA4': PFOA, PFNA, PFOS and PFHxS) have not, to our knowledge, been risk assessed yet and limit values have not been set. It is beyond the scope of this assignment to risk assess these 8 PFAS. There are comparable 28-day experiments on rats for several PFCA and PFSA conducted by NTP in the USA and on the basis of these reports, RIVM has determined 'Relative Potency Factors' for a number of PFAS (publication attached). The results from this may be used to compare the toxicity of the individual PFAS with different chain lengths. However, there is much evidence to suggest that using data from rodents in the risk assessment will underestimate the risk to PFAS in humans. An assessment of this is a comprehensive task that is beyond the scope of the drawing right.

Calculation of PFAS uptake in meat

PFAS content can be estimated using a method where transfer factors are used in the first step to estimate PFAS uptake in serum from e.g. drinking water. We found transfer factors for 5 PFAS from drinking water to serum in cattle (Table 1). In the next step, partition coefficients between serum and tissue (e.g. muscle, which is what we typically eat) are used to estimate the final concentration in the meat. We found partition coefficients for 3 PFAS (Table 2).

The study reporting on transfer factors from contaminated drinking water to cattle is very thorough (Drew et al. 2021). The study has been run over a long period of time, which means that steady state has been reached, and this is important for the reliability of the data. The results for PFAS show the so-called 'transfer factor', i.e. the ratio between the content in serum relative to the content in drinking water in adult cattle that are not pregnant and not dairy cattle. These transfer factors apply if the animals have been exposed to the contaminated water or feed for at least 3-5 half-lives. This corresponds to 222-370 days in cattle (T½=74 days) and is

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This is the case in this study. If the animals have been exposed for a shorter time, the transfer factors will be overestimated.

Table 1. Transfer factors for PFAS from drinking water(Drew et al. 2021)**or silage/barley**(Vestergren et al. 2013)**for serum in cattle**.*Transfer factors for cattle were calculated by dividing the steady state serum concentration (ng PFAS/mL) by the water concentration (ng PFAS/mL).*

Source	Animal	Receive compartment	Transfer factors (average level) (serum conc/water conc)					
			PFHxS	PFHpS	PFOS	PFOA	PFNA	PFDA
Drink-kevand	Adult cattle (non-pregnant, non-lactating gutter)	Serum	65	130	140		170	120
Silage/barley	Adult cattle (non-pregnant, non-lactating gutter)	Serum	-	-	30[a]	~0.4[a]	~12[a]	~10[a]

a: the figures are taken from figure 2 in Vestergren et al. 2013.

In Table 2, partition coefficients for PFOS, PFOA and PFHxS from tissue to plasma are given. Since PFAS levels in serum and in plasma can be set to 1:1, these factors can be used to calculate the PFAS level in muscle tissue and liver at a specific contamination of serum. Data from Kowalczyk et al. 2013 where both PFOS, PFOA and PFHxS were studied indicate that the partition coefficients are relatively comparable for the three substances (0.07, 0.07 and 0.05).

Table 2. Tissue-to-serum partition coefficients for PFAS in cattle(modified from Death et al 2021)

Species	Study	**PFOS**		**PFOA**		**PFHxS**	
		Muscle	Liver	Muscle	Liver	Muscle	Liver
Cattle	*LANUV 2011*[#]	0.07					
	Kowalczyk et al 2013	**0.07**	1.6	**0.07**	1.2	**0.05**	0.16
	Vestergren et al 2013	0.09	0.6[b]	0.2	0.3[c]		
	*Lupton et al 2014***	0.02	0.3				
	*Lupton et al 2015***	0.03	1.0				

[#] Data from three fattened bulls; ** steady-state assumed not to have been achieved;[b]Mean PFOS concentrations in whole blood (converted from ng/kg to ng/L using a specific gravity of 1.06 from Trudnowski and Rico (1974), liver, muscle, and milk were 117 ± 20 (234 ng/L in serum), 130 ± 32, 21 ± 19 ng/kg, and 6.2 ± 1.1 ng/L, respectively.

Calculation methods

PFOS serum concentration in cattle:	2 ng/L x 140	= 280 ng/L	
My PFOS concentration in beef (muscle):	280 ng/L x 0.02	**=5.6 ng/L ~**	**6.0 ng/kg meat**
Max PFOS concentration in beef (muscle):	280 ng/L x 0.09	**=25.2 ng/L~**	**27.1 ng/kg meat**

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Estimation of transfer factors for other PFAS

For other PFAS without data for transfer factors, we have chosen a conservative approach and therefore propose that they be assigned the same values as PFOS, which is the worst-case with a single exception (Table 1).

Uncertainties

The above calculations should only be seen as a rough guideline. In addition to the general methodological uncertainties that will always be present in the studies that have calculated the value of the transfer factors, it must be remembered that the content in the meat reflects the animals' exposure over several months. The content in the animals' feed and drinking water will naturally vary during the time that the animals are in the contaminated areas, and this variation is not seen unless several measurements have been made from start to finish during the period when the animals have grazed on the area. In addition, animals may have different preferences for where in the field they move (and eat) and the contamination in the area will usually hardly be evenly distributed.

Liver analyses as a supplement

Measurements in muscle tissue should be supplemented with measurements in the liver, partly because consumers of this organ will be particularly exposed, and partly because measurements of the liver will lower the limit for when contaminated animals can be identified, even if the content in the muscle tissue may be below the LOQ of the method. In dairy cattle, the PFOS level in muscle (i.e. meat) is 1/100 of the concentration in the liver (Death et al 2021).

References

Death C, Bell C, Champness D, Milne C, Reichman S, Hagen T. Per- and polyfluoroalkyl substances (PFAS) in livestock and game species: A review. Science of The Total Environment, Volume 774, 2021.

Drew R, Hagena TG, Champness D. Accumulation of PFAS by livestock – determination of transfer factors from water to serum for cattle and sheep in Australia. Food Additives & Contaminants: Part A, 2021. DOI: 10.1080/19440049.2021.1942562

Kowalczyk J, Ehlers S, Fürst P, Schafft H, Lahrssen-Wiederholt M. Transfer of Perfluorooctanoic Acid (PFOA) and Perfluorooctane Sulfonate (PFOS) From Contaminated Feed Into Milk and Meat of Sheep: Pilot Study. Arch Environ Contam Toxicol (2012) 63:288–298.

Kowalczyk J, Ehlers S, Oberhausen A, M Tischer, Fürst P, Schafft H, Lahrssen-Wiederholt M. Absorption, Distribution, and Milk Secretion of the Perfluoroalkyl Acids PFBS, PFHxS, PFOS, and PFOA by Dairy Cows Fed Naturally Contaminated Feed. J. Agric. Food Chem. 2013, 61, 12, 2903.

Vestergren R, Orata F, Berger U & Cousins IA. Bioaccumulation of perfluoroalkyl acids in dairy cows in a naturally contaminated environment. Environ Sci Pollut Res (2013) 20:7959–7969.

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Appendix 4: Table S2 of the review article by Death *et al.* (2021)

Table S2. Tissue to serum partition coefficients of per- and polyfluoroalkyl substances (PFAS) in livestock studies (modified from ToxConsult 2016 a,b)

Species	Study	PFOS			PFOA			PFHxS		
		Muscle	Liver	Kidney	Muscle	Liver	Kidney	Muscle	Liver	Kidney
Sheep	Kowalczyk et al 2012*	0.15	5.1	1.1	0.02	0.2	0.4			
Pig	Kowalczyk et al 2014%	0.1	5	1.2	0.06	0.4	0.45			
	Numata et al 2014%							0.05	0.2	0.15
Cattle	LANUV 2011#	0.07								
	Kowalczyk et al 2013	0.07	1.6	0.6	0.07	1.2	1.0	0.05	0.16	0.3[a]
	Vestergren et al 2013	0.09	0.6[b]		0.2	0.3[c]				
	Lupton et al 2014**	0.02	0.3	0.07						
	Lupton et al 2015**	0.03	1.0	0.6[d]						
Chicken	Kowalczyk et al 2014@	0.08	1.1	0.9	0.05	0.6	3			
	Yeung et al 2009^		0.3-0.7	0.2-0.6		0.4	0.9			

See main article for references

* Data from two ewes; steady-state assumed not to have been achieved

% Data are approximate averages for eight gilts, eight barrows and eight boars. There were no marked differences in tissue:plasma ratios for the different groups.

Data from three fattened bulls.

** steady-state assumed not to have been achieved

@ Data from four laying hens. PFOS egg yolk to serum ratio: 8-15; PFOS egg white to serum ratio: 0.004-0.007. PFOA egg yolk to serum ratio: 3.3; PFOA egg white to serum ratio: 0.008

^ Data from 6 day-old male chicks.

[a] Ratio reflects the average of the two sampling times (day 29 at the end of the PFAS-feeding period, and day 44, the end of the PFAS-depuration period).

[b] Mean PFOS concentrations in whole blood (converted from ng/kg to ng/L using a specific gravity of 1.06 from Trudnowski and Rico (1974), liver, muscle, and milk were 117 ± 20 (234 ng/L in serum), 130 ± 32, 21 ± 19 ng/kg, and 6.2 ± 1.1 ng/L, respectively.

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Appendix 5: (i) raw data for PFOS (L-PFOS, m-PFOS and di-PFOS) in muscle tissue and blood plasma from 28 cows as determined in the study by Johnston *et al.* (2023) (Table B5-a), and (ii) Ppartition coefficients (PC) of total PFOS (sum of individual values for L-PFOS, m-PFOS and di-PFOS) between muscle tissue and blood plasma of these animals (Table B5-b)

Table B5-a

Table B*. Paired PFOS Muscle-Plasma Concentrations

		Muscle ng/g at necropsy				Plasma ng/mL at necropsy			
Id Number	Category	L-PFOS*	3Me-PFOS**	6Me-PFOS***	6Me-PFOS Censored	L-PFOS*	3Me-PFOS**	6Me-PFOS***	Days Removed from Exposure at Necropsy
1771	Dry	4.3513	3.4293	1.5485	0	49.4900	34.4700	29.9150	14
21634	Lactating	4.8649	4.2329	1.0697	0	66.6150	45.6900	36.9200	14
22242	Lactating	3.8373	2.9174	1.1248	0	41.2000	45.5850	30.0150	14
23999	Lactating	4.3206	2.6816	1.5154	0	27.9600	32.0100	18.6800	14
29298	Lactating	2.0231	0.6295	0.3453	0	11.0500	10.8450	7.0750	14
29839	Lactating	3.3338	1.7090	1.0364	0	33.9850	47.1800	27.1150	14
31230	Dry	3.0129	2.1008	1.0765	0	41.0200	50.3900	31.3250	14
31278	Lactating	4.2018	1.9909	1.1999	0	41.6400	29.2450	23.1750	14
34526	Lactating	3.3916	1.5254	0.9540	0	26.9750	33.7350	20.8700	14
36567	Dry	3.3183	2.2940	1.2692	0	43.3450	26.3050	23.4800	14
37422	Dry	3.9829	2.9525	1.6847	0	52.2400	64.5150	39.5350	14
37808	Lactating	2.6556	1.3173	0.8001	0	30.9250	43.6650	25.7450	14
38528	Lactating	3.6013	1.7132	1.0030	0	39.6650	46.8300	28.1400	14
39140	Dry	2.9939	1.2166	0.8681	0	37.1250	24.4350	23.6550	14
39627	Lactating	1.3779	0.5825	0.4014	0	33.7800	17.9200	17.7600	14
40811	Young	2.2049	0.9716	0.6794	0	47.1100	26.0400	25.3650	14
40956	Young	1.2437	0.5524	0.3121	0	16.6400	21.9750	14.0200	14
40964	Young	1.4447	0.5688	0.3457	0	31.8000	17.1650	16.6250	14
41061	Young	1.3055	0.5685	0.3171	0	15.8450	20.6300	14.1150	14
41301	Young	1.0093	0.3822	0.2411	1	13.6950	8.2450	7.3300	14

Method detecti quantification limit	
ng/g	LOD*
L-PFOS	0.168
3Me-PFOS	0.070
6Me-PFOS	0.182
LOD*: level of detect	
LOQ**: level of quan	

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Table B5-a (continued)

1833	Lactating	0.8571	0.5322	0.4039	0	10.3400	9.3650	7.3650	153
33866	Lactating	0.6017	0.4349	0.2211	1	15.0200	13.9650	10.9650	137
38679	Lactating	1.4243	0.8242	0.5278	0	24.6950	19.5100	16.8800	153
40807	Young	0.9550	0.5164	0.2648	0	17.2250	11.2150	11.2800	153
40875	Young	0.7548	0.5220	0.2779	0	15.9550	10.2050	10.6950	137
41034	Young	0.4569	0.1346	0.1751	1	6.7400	4.2450	4.2200	153
41086	Young	0.5549	0.2525	0.2609	0	6.9200	4.9000	4.6200	137
41366	Young	0.7579	0.2237	0.2384	1	11.7600	6.6700	6.8650	137

L-PFOS*: linear isomer
3Me-PFOS**: branched isomer
6Me-PFOS***: branched isomer

Table B5-b

Identification number	Category	Number of days of retention of exposure to PFOS at slaughter	PFOS conc. muscle tissue (µg/kg)	PFOS conc. blood plasma (µg/L)	PC (conc. PFOS in meat/conc. PFOS in blood plasma
1771	dry	14	9.33	113.75	0.08
21634	lactating	14	10.17	149.23	0.07
22242	lactating	14	7.88	116.80	0.07
23999	lactating	14	8.52	78.65	0.11
29298	lactating	14	3.00	28.97	0.10
29839	lactating	14	6.08	108.28	0.06
31230	dry	14	6.19	122.74	0.05
31278	lactating	14	7.39	94.06	0.08
34526	lactating	14	5.87	81.58	0.07
36567	dry	14	6.88	93.13	0.07
37422	dry	14	8.62	156.29	0.06
37808	lactating	14	4.77	100.34	0.05
38528	lactating	14	6.32	114.64	0.06
39140	dry	14	5.08	85.22	0.06
39627	lactating	14	2.36	69.46	0.03

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40811	heifer	14	3.86	98.52	0.04
40956	heifer	14	2.11	52.64	0.04
40964	heifer	14	2.36	65.59	0.04
41061	heifer	14	2.19	50.59	0.04
41301	heifer	14	1.63	29.27	0.06
1833	lactating	153	1.79	27.07	0.07
33866	lactating	137	1.26	39.95	0.03
38679	lactating	153	2.78	61.09	0.05
40807	heifer	153	1.74	39.72	0.04
40875	heifer	137	1.55	36.86	0.04
41034	heifer	153	0.77	15.21	0.05
41086	heifer	137	1.07	16.44	0.06
41366	heifer	137	1.22	25.30	0.05

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Appendix 6: Results of analyses of PFOS, PFOA, PFNA and PFHxS in blood serum (unit = µg/L), meat and liver of bovine animals as carried out by the FASFC on samples taken in the period July 2023 to July 2024

Analyses NC PFAS in bovine meat

Animal	Herd	Age	Sex	Earmark	Butchered	Sample N°	PFOS (µg/kg)	PFOA (µg/kg)	PFNA (µg/kg)	PFHxS (µg/kg)	4PFAS (µg/kg)
Dairy cow	Herd 1	4 years	F	bovine 8	23/10/2023		0.53	<0,05	<0,05	<0,05	0.53
Dairy cow	Herd 2	7 years	F	bovine 9	25/10/2023		5	<0,05	<0,05	<0,05	5
Beef cow	Herd 3	3,5 years	F	bovine 10	11/07/2023		1.2	<0,02	<0,02	<0,02	1.2

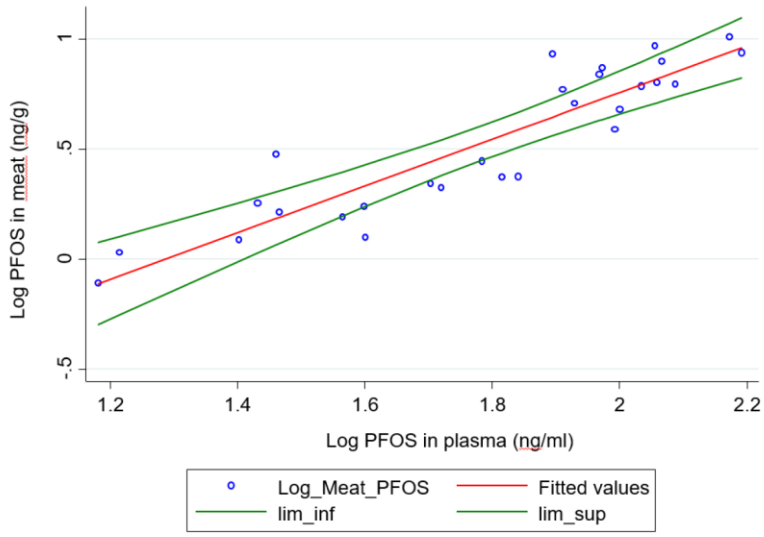
---> euthanised animal not intended for consumption

Analyses PFAS in the meat, liver and blood of bovines from contaminated herds

Animal	Herd	Age	Sex	Earmark	Butchered	Sample N°	PFOS (µg/kg)			PFOA (µg/kg)			PFNA (µg/kg)			PFHxS (µg/kg)			4PFAS (µg/kg)			Measurement uncertainty	PFOS blood/PFOS meat	Remarks
							Meat	Liver	Blood	Meat	Liver	Blood	Meat	Liver	Blood	Meat	Liver	Blood	Meat	Liver	Blood			
EU norms							0.3	6	-	0.8	0.7	-	0.2	0.4	-	0.2	0.5	-	1.3	8	-			
Dairy cow	Herd 2	5 years	F	bovine 1	14/12/2023		3			< 0,050			< 0,050			< 0,050			3			30%		
								55			< 0,1			< 0,1			< 0,1			55		30%	21.0	
									63			< 0,1			< 0,1			0.51			64	PFOS : 52%, PFHxS : 30%		
Dairy cow	Herd 2	6 years	F	bovine 2	14/12/2023		2.2			< 0,050			< 0,050			< 0,050			2.2			30%		
								55			< 0,1			< 0,1			< 0,1			55		30%	24.5	
									54			< 0,1			< 0,1			0.39			54	PFOS : 52%, PFHxS : 30%		
Dairy cow	Herd 1	6 years	F	bovine 3	03/04/2024		0.3			< 0,050			< 0,050			< 0,050			0.3			30%		
								6.8			< 0,05			< 0,05			< 0,05			6.8		30%	13.3	
									4			< 0,1			< 0,2			0.14			4.1	PFOS : 52%, PFHxS : 30%		
Bull	Herd 1	2,5 years	M	bovine 4	03/04/2024		0.56			< 0,050			< 0,050			< 0,050			1.7			30%		only animal purchased by XXXXXXX
								19			< 0,05			< 0,05			< 0,05			19		30%	10.9	
									6.1			< 0,1			< 0,2			< 0,1			6.1	PFOS : 52%, PFHxS : 30%		
Dairy cow	Herd 2	8 years	F	bovine 5	28/05/2024		3			< 0,050			< 0,050			< 0,050			3					3 calvings
								86			< 0,1			< 0,1			< 0,1			86			26.0	
									78			< 0,1			< 0,2			0.55			79			
Dairy cow	Herd 2	8 years	F	bovine 6	28/05/2024		3.5			< 0,050			< 0,050			< 0,050			3.5					5 calvings
								93			< 0,1			< 0,1			< 0,1			93			24.9	
									87			< 0,1			< 0,2			0.35			87			
Dairy cow	Herd 2	6,5 years	F	bovine 7	28/05/2024		3.9			< 0,050			< 0,050			< 0,050			3.9					2 calvings
								84			< 0,1			< 0,1			< 0,1			84			20.0	
									78			< 0,1			< 0,2			0.65			79			

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Appendix 7: Linear regression analysis applied to PFOS levels in blood plasma and muscle tissue of the same animals (paired results) from the study by Johnston *et al.* (2023) (see Appendix 5). Based on the regression parameters, a PFOS concentration in bovine blood corresponding to the European maximum level (Regulation (EU) 2023/915) of PFOS in bovine meat was determined with a confidence level of 95%. Stata statistical software was used for the regression analysis.




```
. xi: regr log_meat_pfos log_plasma_pfos
```

Source	SS	df	MS		
				Number of obs	= 28
				F(1, 26)	= 129.49
Model	2.46923116	1	2.46923116	Prob > F	= 0.0000
Residual	.495783241	26	.019068586	R-squared	= 0.8328
				Adj R-squared	= 0.8264
Total	2.9650144	27	.109815348	Root MSE	= .13809

log_meat_pfos	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
log_plasma_pfos	1.05839	.0930088	11.38	0.000	.8672078	1.249573
_cons	-1.363653	.1685613	-8.09	0.000	-1.710136	-1.01717

Determination of a blood plasma level of PFOS that corresponds to a PFOS level in meat according to the European maximum level of 0,3 µg/kg of meat:

- log (0,3 µg/kg PFOS) = - 0,5299;
- according to the regression parameters, this corresponds to a log-value of 0,7944 for blood plasma, with a 95% confidence interval with 0,3956 and 1,3691 as lower and upper limits, respectively;
- after transformation, a blood plasma level of 6,2286 µg PFOS/l is obtained, with a 95% confidence interval with 2,4864 and 23,3915 as lower and upper limits, respectively;
- the coefficient of determination is 0,8264 and the *P*-value is lower than 0,0001;
- the uncertainty is due to the limited size of the data set (n = 28) and the low number of data with a low level of contamination;
- note that the density of blood plasma is equal to 1,024 kg/m^3.

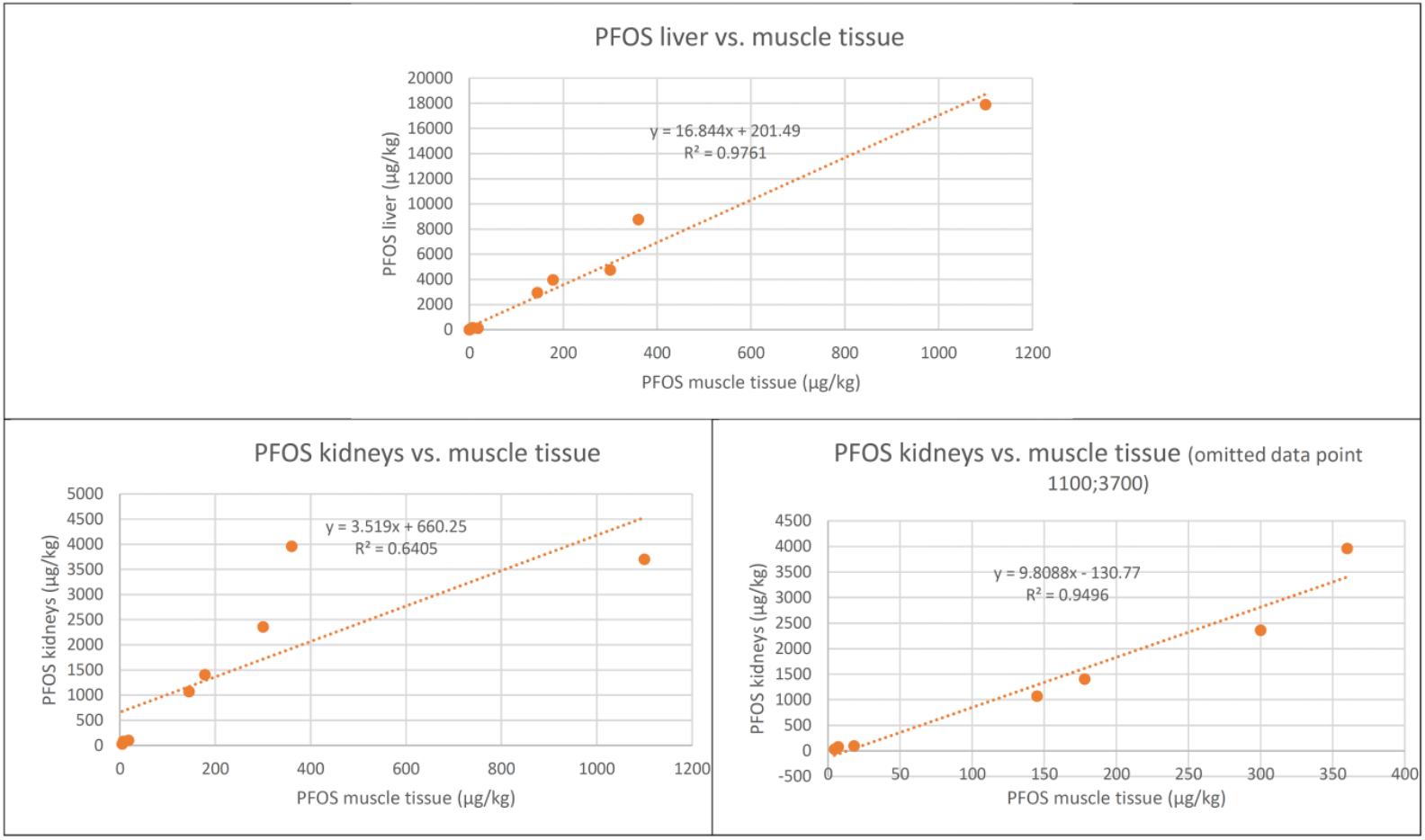
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Appendix 8: Collected data on PFAS concentrations in bovine muscle tissue, liver and kidneys, and calculation of liver/muscle tissue and kidney/muscle tissue concentration ratios, where M, L and K mean meat, liver and kidneys, respectively

		PFOS (µg/kg)					PFOA (µg/kg)					PFNA (µg/kg)			PFHxS (µg/kg)					Source
		M	L	L/M	K	K/M	M	L	L/M	K	K/M	M	L	K	M	L	L/M	K	K/M	
	Norm	0.3	6		6		0.8	0.7		0.7		0.2	0.4	0.4	0.2	0.5		0.5		
Animal	Age (years) or weight																			
dairy cow	5	3	55	18			< 0,050	< 0,1	/			< 0,050	< 0,1		< 0,050	< 0,1	/			FASFC, appendix 6
dairy cow	6	2.2	55	25			< 0,050	< 0,1	/			< 0,050	< 0,1		< 0,050	< 0,1	/			FASFC, appendix 6
dairy cow	7	0.3	6.8	23			< 0,050	< 0,05	/			< 0,050	< 0,05		< 0,050	< 0,05	/			FASFC, appendix 6
dairy cow	8	3	86	29			< 0,050	< 0,1	/			< 0,050	< 0,1		< 0,050	< 0,1	/			FASFC, appendix 6
dairy cow	8	3.5	93	27			< 0,050	< 0,1	/			< 0,050	< 0,1		< 0,050	< 0,1	/			FASFC, appendix 6
dairy cow	6.5	3.9	84	22			< 0,050	< 0,1	/			< 0,050	< 0,1		< 0,050	< 0,1	/			FASFC, appendix 6
		mean		24																
		standard deviation		4																
		minimum		18																
		median		24																
		maximum		29																
bull	2.5	0.56	19	34			< 0,050	< 0,05	/			< 0,050	< 0,05		< 0,050	< 0,05	/			FASFC, appendix 6
dairy cow (n=3), group 1	583 kg	145	2952	20	1074	7	0.6	10.1	17	8.7	15				19.1	60.8	3	98.2	5	Kowalczyk *et al.* (2013)
standard deviation		*36*	*718*		*153*		*0.3*	*1.9*		*3.9*					*7.1*	*23.7*		*30.8*		Kowalczyk *et al.* (2013)
dairy cow (n=3), group 2	583 kg	178	3964	22	1408	8	< 0,2	0.8	/	0.4	/				4.9	18.6	4	39.4	8	Kowalczyk *et al.* (2013)
standard deviation		*48*	*1035*		*303*		/	*0.7*		/					*2.7*	*9.6*		*22.4*		Kowalczyk *et al.* (2013)
dairy cow (n=5)	29 tot 45 months	0.021	0.13	6			0.007	0.009	1			0.0045	0.016		< LOD	< LOD	/			Vestergren *et al.* (2013)
standard deviation		*0.02*	*0.032*				*0.0043*	*0.002*				*0.0033*	*0.0034*							
bull (n=3)	329 kg	1100	17900	16	3700	3														Lupton *et al.* (2014)
standard deviation		*100*	*2300*		*500*															
bull (n=2) - 343 days	307 kg	7	149	21	82	12														Lupton *et al.* (2015)
standard deviation		*1*	*15*		*24*															
dairy cow (n=2) - 105 days	308 kg	360	8760	24	3960	11														Lupton *et al.* (2015)
standard deviation		*40*	*3540*		*160*															
dairy cow (n=2) - 343 days	308 kg	300	4740	16	2360	8														Lupton *et al.* (2015)
standard deviation		*10*	*560*		*570*															
bovines ("cattle") (n=12) 63 days	1,4-12,3	18	120	7	100	6														Drew *et al.* (2021) (values derived from figures)
heifers (n=5) - 215 days	1,4-2	4.5	100	22	32	7														Drew *et al.* (2021) (values derived from figures)
		mean		21		8														
		standard deviation		7		3														
		minimum		6		3														
		median		22		8														
		maximum		34		12														
		n		16		8														

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Appendix 9: Application of linear regression to the PFOS levels in bovine muscle tissue, liver and kidneys from Table 4. In particular, linear regression is used to obtain an equation regarding the correlation between (i) PFOS concentrations in muscle tissue and liver of the same animals mutually and (ii) PFOS concentrations in muscle tissue and kidneys of the same animals mutually. Care should be taken to apply the obtained equations in practice, as the number of data points is rather limited.



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Appendix 10: Summary of studies determining elimination half-lives of PFOS, PFOA, PFNA and/or PFHxS in bovine blood. The table is partly based on the overview table from Mikkonen *et al.* 2013

Reference	Race, sex and lactation status	Age and/or weight	Method and frequency of PFAS exposure	Degree of exposure to PFAS	Frequency of sampling	Reported (mean) elimination half-lives in blood (plasma/serum)
Lupton *et al.* (2012)	Lowline Angus steers	329 ± 36 kg	a single oral bolus	1 mg/kg body weight of ^{14}C-PFOA	0, 1, 2, 4, 8, 12, 24, 36, 48, 72 and 96 hours, and on day 6, 8, 10, 13, 15, 17, 21, 24 and 28	19.2 ± 3.3 days (n = 4) for PFOA in plasma
van Asselt *et al.* (2013)	lactating cows (breed unknown)	2.5-5.5 years old	fed with naturally contaminated feed (hay and silage) for 28 days	5.3-9.6 µg PFOS/kg body weight/day	Not stated	estimated half-life of 56 days for PFOS
Vestergren *et al.* (2013)	lactating Swedish Red cows	> 2 years	chronic environmental exposure	estimated exposure of 0.00052 µg PFAS/kg body weight/day	Not stated	38.7 days for PFOS, 1.3 days for PFOA, and 8.7 days for PFNA in (whole) blood
Lupton *et al.* (2015)	Angus steers and heifers	~ 343 kg	single oral bolus dose (gelatin capsule)	steers: 0.098 mg PFOS/kg body weight; fears: 9.09 mg PFOS/kg body weight	0, 4, 12, and 24 hours, and on day 2, 3, 6, 9, 15, 22, 30, 42, 56, 77, 105, 119, 133, 161, 175, 203, 238, 266, 294, 329, and 343	steers: 120 ± 4.1 days (n = 4) for PFOS in plasma; heifers: 106 ± 23.1 days (n = 4) for PFOS in plasma
Drew *et al.* (2021)	Belted Galloway bovines (beef cattle)	1.4-12.3 years old	chronic environmental exposure	stock water with approx. 12.3 µg/LPFAS	5 time points between day 19 and 120 for 5 animals (1.4-2 years), supplemented by additional data from 11 animals (1.4-12.3 years) at two time points	74.1 ± 13.4 days (n = 16) for serum PFOS; 45.7 ± 9 days for PFHpS[a] in serum; 9.3 ± 1.3 days (n = 16) for serum PFHxS; 12.3 ± 3.2 days for PFNA in serum; 60.4 ± 10.4 days for PFDA[b] in serum

[a] PFHpS = perfluoroheptanesulfonic acid; [b] PFDA = perfluorodecanoic acid

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